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IŞIK UNIVERSITY
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NATURAL SCIENCES

MECHANICAL ENGINEERING DEPARTMENT

MECH/MECT4902

GRADUATION DESIGN PROJECT

LINK DRIVE MECHANICAL PRESS DESIGN FOR AUTOMOTIVE INDUSTRY

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Preface

In this graduation design project, a mechanical press system was first designed to meet industrial requirements for metal forming. Then, a complete link drive mechanism was developed, including the main gear, eccentric gear, and ram assembly. Using appropriate material selection, realistic loading, and actual boundary conditions to simulate real-world machine.

We owe our sincere thanks to our project advisor Assist. Prof. Dr. Umut Karagüzel, due to the dedication he showed during our project study, the support, his guidance to us with his experience and knowledge during our work.

In addition, we would like to thank Prof. Dr. Mehmet Demirkol, Assist. Prof. Dr. Ali Taner Kuzu, Assist. Prof. Dr. Onur Keskin, Assist. Prof. Dr. Sinan Eren Yalçın and Assoc. Prof. Erkin Dinçmen, Prof. Olcay Türkoğlu for who did spare their help and support during our education.

Finally, and most significantly, we would like to thank our families, who have never given up their support.

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Abstract

This project is all about designing and analyzing a high-performance Link Drive Mechanical Press for the automotive industry, where you really need strong force, precise movement, and durability. The press can handle 300 tons, have a stroke length of 130 mm, and work at 15 strokes per minute. The main goal here is to tweak the motion of the linkage to get the most time at the Top Dead Centre (TDC), which is super important for shaping metal properly. We did a thorough look at how the system works, using MATLAB to fit stroke profiles into polynomials and figuring out the speed and acceleration from that. The stroke function we came up with improves timing and dwelling, which is a big plus for sheet metal forming. We also modelled key parts like the main gear, eccentric gear, connecting rod, and ram to check for stress using basic mechanics. We paid special attention to gear loading, torsion in the shafts, bending from the eccentric setup, and stress conditions while simulating them in Ansys FEA for ensuring safety. We picked materials carefully to handle high loads., Maraging Steel 300, and AISI 4340 came out on top for different structural elements because of their yield strength, resistance to wear, ability to handle heat, and protection from rust. We kept the Safety Factors (FoS) above 1.7 for all parts under maximum load to ensure they're strong enough.

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1- Introduction

We selected a link-drive mechanical press due to its non-uniform slide motion, which offers faster approach and return strokes, and slower forming speed near bottom dead center (BDC), the critical phase where peak force and die engagement occur. This optimized kinematic profile minimizes shock loads, improves forming accuracy, and extends die life. This behavior is well-explained through four-bar and slider-crank linkage principles.

We deliberately ruled out other press types based on performance, efficiency, and application limitations:

Crank (eccentric) presses: while mechanically simpler, they produce uniform slide velocity, which leads to high impact force at BDC, reduced control during forming, and higher tool wear. This uniform motion limits their effectiveness in deep drawing and fine blanking operations

Hydraulic presses: offer precise force control and full-tonnage capability throughout the stroke. However, they are inherently slower, less energy-efficient, and involve higher maintenance due to fluid systems. This makes them less suitable for high-volume, fast-cycle production in automotive settings.

Our link-drive press uniquely balances speed, force, and control. Its kinematic profile allows for slower slide velocity near BDC, reducing defects in sheet forming and allowing complex geometries and tight tolerances — critical in automotive stamping, such as body panel and structural component production. With its mechanical efficiency, lower die wear, and suitability for automated production lines, it is the most effective solution for modern manufacturing demands.

1.1. Project Objective

The primary goal of this project is to design and evaluate a 300-tonne link drive mechanical press with optimized stroke characteristics for the automotive industry. The machine must:

- Deliver 130 mm of stroke
- Operate at 15 strokes per minute

- Provide a custom stroke profile with extended dwell at TDC
- Maintain high structural integrity under cyclic high-load conditions

This involves in-depth study and implementation of linkage kinematics, force transmission analysis, stress calculations, and material optimization for each critical component.

Scope of Work

The following steps were considered in our project:

Stroke Profile Optimization: The stroke path was experimentally and computationally optimized using MATLAB, producing an 8th-degree polynomial fit of stroke vs. crank angle, allowing easy derivation of velocity and acceleration curves. The green curve, among several iterations, was selected for maximizing TDC dwell.

Kinematic Analysis: A four-bar linkage system was analyzed under forward and reverse operation using simulation tools. Displacements, velocities, and accelerations were computed for all moving components, enabling the identification of optimal operating conditions and geometry. (GEOGEBRA)

Stress and Structural Analysis: Classical and technical mechanics methods were used to evaluate key stresses:

1. Gear tooth stress using Lewis and Hertzian formulas
2. Shaft torsional and bending stress
3. Eccentric loading of gears
4. FEA Analysis

Combined axial, bending, and shear stress in rods and arms

Material Selection: For each component, material choices were guided by performance under stress, machinability, cost, and fatigue behaviour. Materials such as Maraging Steel 300 and AISI 4340 ensure mechanical resilience and operational longevity.

Design Validation: Safety factors exceeding 1.7 were achieved in all areas, confirming robustness. The analysis accounted for the worst-case loading scenarios, ensuring reliable operation under industrial conditions.

2- Design Constraints

Technical Constraints:

- The press must deliver a maximum force capacity of 3000 kN (300 tons) with a stroke of 130 mm operating at 15 strokes per minute, imposing strict structural and kinematic requirements on the mechanism.
- The stroke profile must provide prolonged dwell time at Top Dead Center (TDC) to accommodate complex metal forming operations typical in automotive applications.
- All components are required to maintain structural integrity under cyclic loading, with a minimum Factor of Safety (FoS) of 1.7, guiding material selection and geometric design.
- Gear and linkage dimensions such as the main gear diameter (500 mm) and number of teeth (20) are fixed, limiting the allowable design envelope.
- Power and torque requirements (e.g., 107,150 Nm torque, 168.2 kW power) must be met without overloading the drive system or increasing energy consumption beyond practical limits.

Operational and Practical Limitations

- The system must fit with conventional industrial press frameworks, ensuring compatibility with existing automotive production lines.
- Components such as gears, shafts, and linkages must be serviceable and replaceable, with design considerations for accessibility, alignment, and lubrication.
- The overall design must ensure low maintenance requirements, high durability, and safe operation under prolonged cyclic loading environments.

2.1. Design Constraints Table

Our link drive mechanical press design constraints shown in table (1) below.

Table 1: Design constraints

Constraint Type	Constraint	Specification / Value
Functional Constraints	Pressing Capacity	300 tons (3000 kN)
	Stroke Length	140 mm
	Stroke Rate	15 strokes per minute
	Stroke Profile	Custom – Linkage based, 8th-order polynomial fit
Max Price	80,000 $	

3- Workload (Gantt Chart)

Table 2: Gantt Chart

Task	Responsible Person	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Analyzing The Link Drive Mechanical Press	Everyone														
Making Design Criteria and Determining the Constraints	Everyone														
Kinematic Analysis for Link Drive	Yağız														
Making Concept Design	Yağız and Ahmed														
Flywheel Design	Yağız and Ahmed														
Gear Design	Yağız and Ahmed														
Pres Body Design	Yağız														
Tool Design	Yağız														
Constraint Calculations for Parts (Main gear, Manin shaft etc.)	Ahmed														
Safety calculations	Ahmed														
Material Selection	Ahmed and Baran														
Motor Selection	Mohamed and Baran														
FEA Analysis	Mohamed														
Technical Drawings	Mohamed														
Preparing thesis	Mohamed, Ahmed, Yağız														

4- Concept Design

4.1. Hand sketched Link drive press

Here is the design of Link drive mechanical press using hand sketching shown in figure (1).

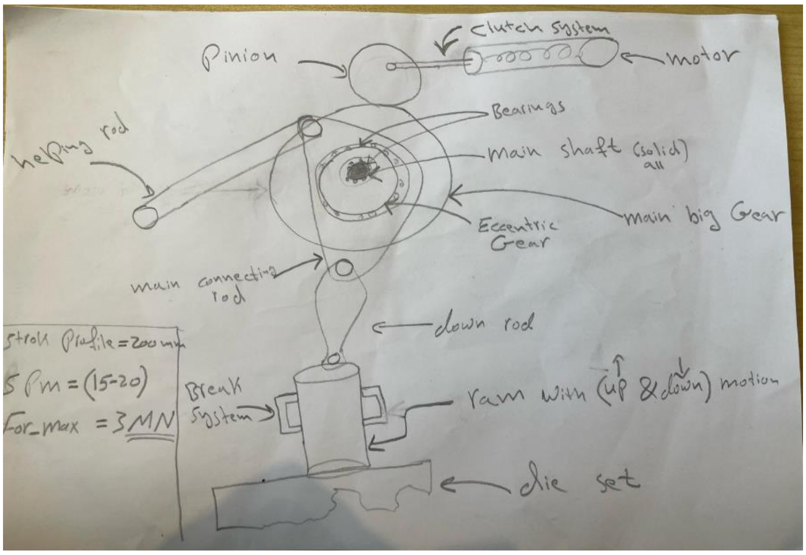

Figure 1: Concept design for link drive mechanical press

4.2. Design of link drive mechanical press

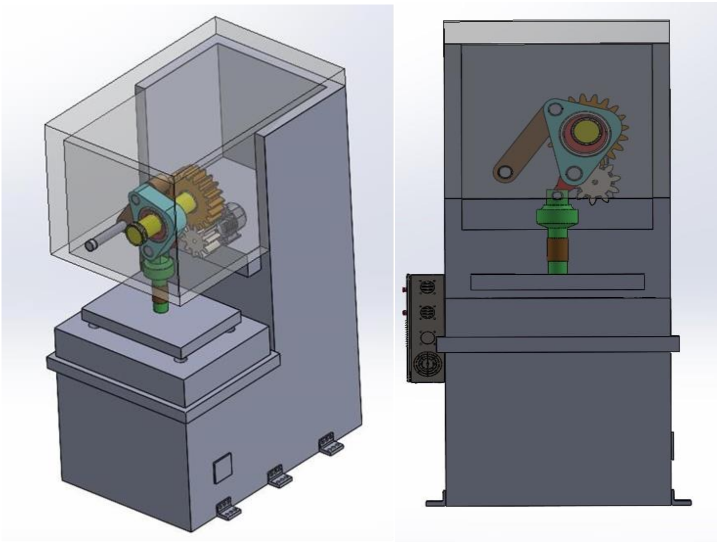

Figure 2: Link drive press design (using CAD)

4.3 Design of Internal Parts

4.3.1 Design of main gear

Main Gear transfers motion to the system when the pinion rotates it. We selected a pitch diameter of 500 mm as a result of space constraints mechanically, machinability, and to make sure that the gear can withstand the torque as requested without too much deflection or failing.

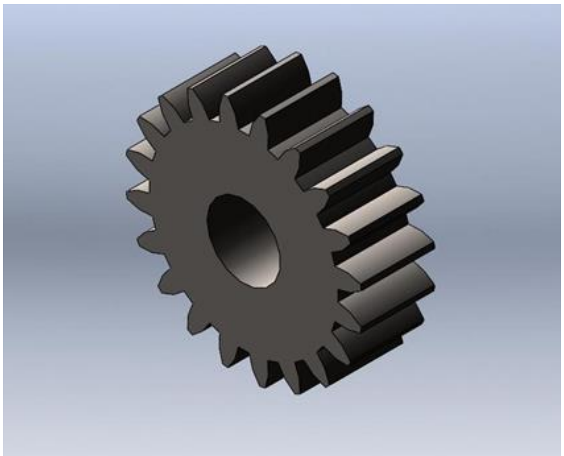

Figure 3: Main gear design

The gear module (m) defines the size of each tooth and is calculated as:

$$m = \frac{D}{Z} = \frac{500}{20} = 25\,mm$$

The gear module is a fundamental metric in metric gear design, representing the ratio between the pitch diameter and the number of teeth. It determines the size and spacing of each tooth, affecting the strength and smoothness of meshing.

The circular pitch (P)

$$(p) = m \times \pi = 78.54 \text{ mm}$$

The circular pitch is basically the distance between two points on neighboring gear teeth, measured along the pitch circle. It's important for how the teeth are spaced and lined up.

4.3.2. Design of main gear shaft

The main gear shaft supports and transmits rotational motion from the main gear to

Eccentric gear.

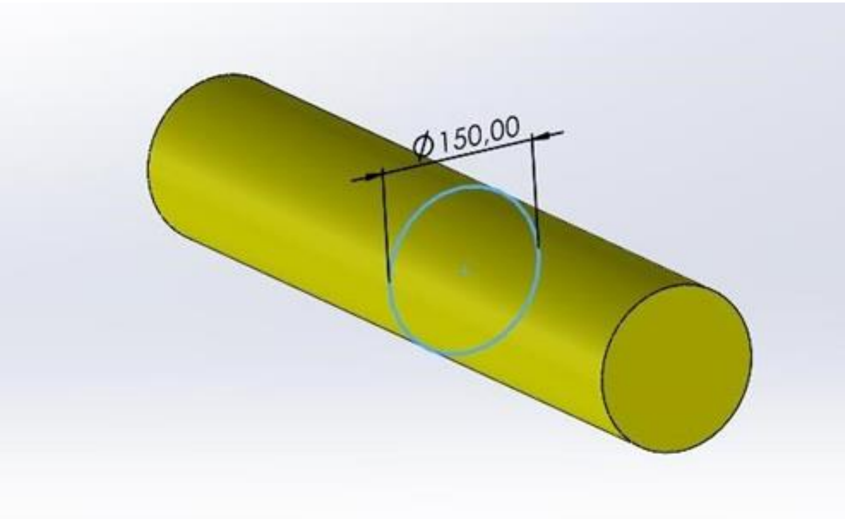

Figure 4: Main gear shaft's design

Shaft dimensions are selected to minimize stress while balancing weight and cost. A larger diameter reduces stress but increases mass. The chosen dimensions aim for optimal performance under the calculated load while ensuring durability and fatigue resistance.

L = 0.5 m

D = 0.15 m

$$\mathbf{Z} = \frac{C}{I} = \frac{D/2}{\pi \times D^4/64} = \frac{32}{\pi \times D^3}$$

The current design works well mechanically, handling both static loads and fatigue while keeping the structure strong for its entire lifespan according to our calculations and analysis.

4.3.3. Design of eccentric gear

Eccentric gear provides the desired stroke motion.

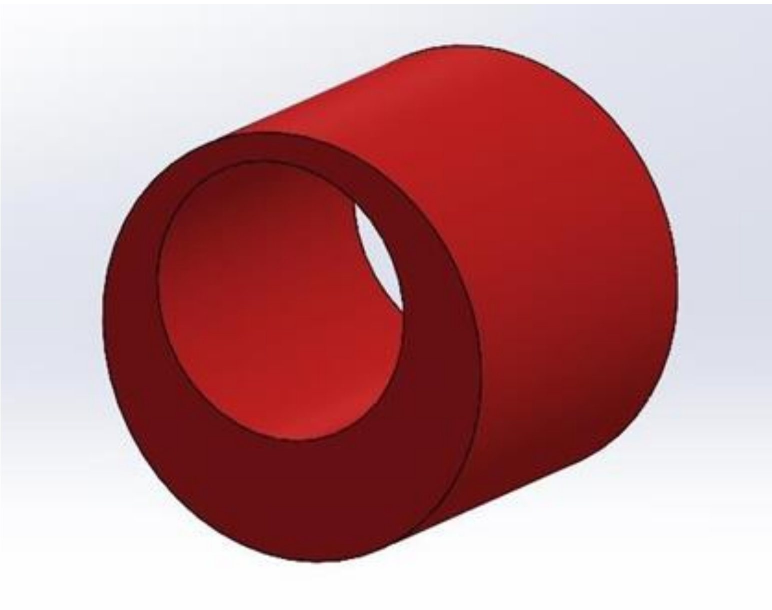

Figure 5: Eccentric gear design

The eccentric gear is a critical component in the link drive system, responsible for converting rotational motion from the main gear into the vertical linear motion of the ram. In our design, the gear has an outer diameter of 400 mm and a total thickness of 300 mm, making it capable of handling the high torque and bending stresses generated during press operation. A key design feature is the 140 mm offset between the shaft bore center and the gear center, which produces the required eccentric motion. The gear is mounted on the main shaft through a 90 mm diameter central bore and is locked in position using a keyway to ensure stable torque transmission without slippage.

4.3.4. Design of connecting rod

Connecting rod links the crank mechanism to the ram to convert rotational motion into reciprocating motion.

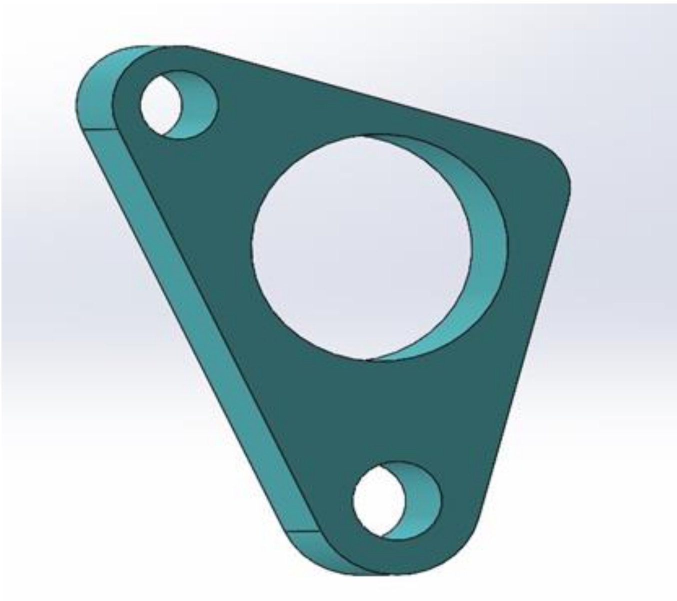

Figure 6: connecting rod design

The connecting rod showing in figure (6), is 350 mm long with a main body height of 150 mm. One end includes a 120 mm diameter bore to house the shaft. The other end is a triangular flange with three mounting holes, each 17 mm in diameter, used for bolting the rod securely to the adjacent part. The overall design ensures that the rod can handle direct axial loading without bending or rotation during press operation. Every dimension was selected based on load path, mounting needs, and machining practicality. No unnecessary features were added.

4.3.5. Design of the upper helping rod

Upper helping arm helps in controlling the stroke motion and distributes force in the system as shown in figure (6).

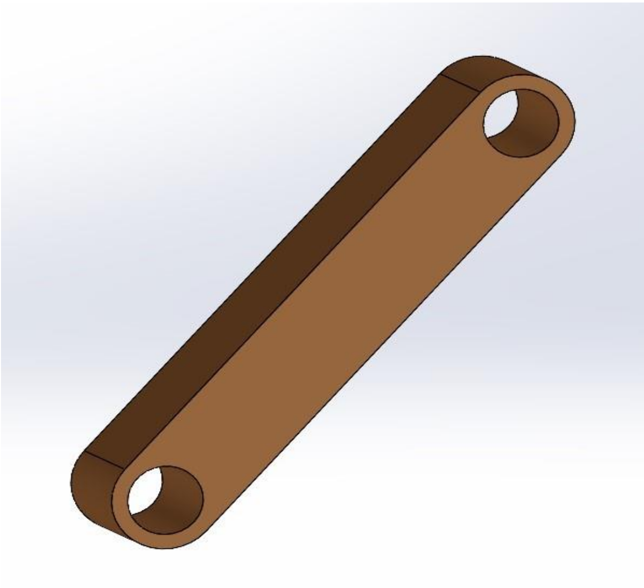

Figure 7:upper helping rod design

It plays a critical role in ensuring the accuracy of motion transmission within the link drive mechanism. Unlike the main load-bearing parts, this rod mainly balances motion and transfers guiding and reaction forces, rather than the full operational load. Because of this, the key design considerations for the upper helping rod are:

- Correct angular motion (trajectory)
- Suitable rod length and thickness for rigidity
- Low deformation to ensure system precision

In a real mechanical system, the helping rod moves between two fixed points. Thus, the actual working angle (θ) must be calculated based on horizontal and vertical distances. Using basic trigonometry:

$$\theta = \arctan\left(\frac{Y}{X}\right) = \arctan\left(\frac{150}{100}\right) = 56.3°$$

- Horizontal distance: X = 100 mm

- Vertical distance: Y = 150 mm

The real angle represents the effective motion path angle of the helping rod between its two connection points.

4.3.6. Design of down rod

Down Arm works with the upper arm to refine the press motion and ensure smooth operation. Shown in figure (8) below.

Figure 8: down rod design

The outer diameter D = 150 mm, the inner diameter d = 80 mm and the thickness t = 150 mm. Which ensures that stresses will be rightly obtained and calculated to maximize the efficiency of the down rod.

4.3.7. Design of the Ram

Ram moves vertically backwards and forwards to apply force on the metal sheet.

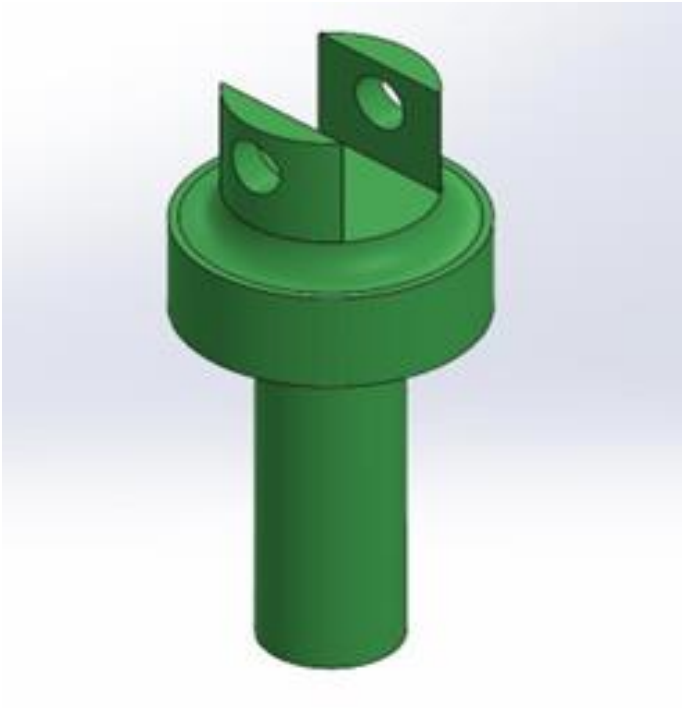

Figure 9: RAM design

The ram is designed as a vertically oriented cylindrical component with a total height of 390 mm. The main shaft section has a diameter of 110 mm and a working length of 300 mm, allowing direct vertical force transmission to the lower press plate. At the top, the ram includes a widened collar section with a diameter of 226 mm and a height of 90 mm to provide increased surface contact and structural reinforcement. This region ensures secure alignment within the press frame and absorbs bending loads. A centered through-hole with a 50 mm diameter is machined at the top end for pin connection to the down rod. The entire design is symmetrical and robust, ensuring minimal deflection under the full 3,000,000 N axial load during press operation.

4.3.8. Design of the Pinion

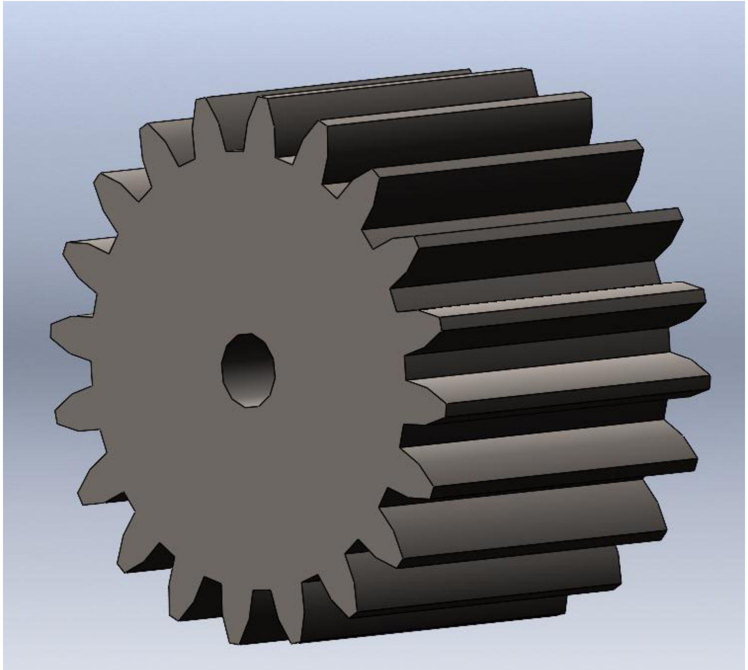

Figure 10: Pinion design

The pinion meshes with the main gear. Therefore, the modulus of it is fixed to match the analyses done before where $m_p = 2.5\ mm$ where the tooth profile is safely fixed to be 20 teeth for pinion with standard involute.

The pinion is smaller than the main gear, but at the same time it will transmit a huge amount of torque:

5- Materials Selection

5.1. Material selection of the main internal parts

Main Gear:

AISI 4340 (Nickel-Chromium-Molybdenum Steel), was selected due to its high yield strength (≈850 MPa), toughness, and excellent wear resistance. This gear experiences high tangential force (428,600 N) and must transfer torque without failing at the teeth or bore. Its high fatigue resistance ensures reliability under cyclical loading during press operations. So, it made it a strong candidate for our main gear material.

Table 3: Pugh matrix for main gear material

Criterion	Weight	AISI 4340	AISI 1045
Yield Strength	3	+1	-1
Fatigue Resistance	3	+1	+0
Wear Resistance	2	+1	-1
Cost	1	+0	+1
Machinability	1	+0	+1
Weldability	1	-1	+1
TOTAL		+7	-2

Main Shaft:

Maraging Steel 300 offers ultra-high strength (Yield ≈1860 MPa) with excellent fatigue and impact resistance, making it ideal for the main shaft, which is subjected to 3 MN axial load, 107,150 Nm torque, and 420,000 Nm bending. It ensures minimum deformation under extreme conditions, essential for precision in press operation.

Table 4:Pugh Matrix for Main Shaft

Criterion	Weight	Maraging 300	AISI 4340
Yield Strength	3	+1	+0
Fatigue Resistance	3	+1	+1
Toughness	2	+1	+1
Cost	1	-1	+0
Weldability	1	-1	+0
TOTAL		+6	+5

Eccentric Gear:

AISI 4340 (Nickel-Chromium-Molybdenum Steel), was chosen for the eccentric gear due to its balance of strength, toughness, and machinability. The part endures both bending and torsional stress due to its 140 mm eccentricity and transmits up to 3 MN axial force. The material's 740–850 MPa yield strength ensures the gear stays within safe stress limits under all loading scenarios.

Table 5:Pugh Matrix for Eccentric Gear

Criterion	Weight	AISI 4340	AISI 4140
Strength	3	+1	+0
Fatigue Resistance	3	+1	+0
Heat-Treatment Capability	2	+1	+1
Availability	1	+1	+1
Cost	1	+0	+1
TOTAL		+9	+4

Connecting Rod & Upper Helping Rod:

AISI 4140 is used due to its moderate strength (Sy ≈860 MPa), good fatigue resistance, and excellent machinability. Both rods undergo combined axial and bending stress, and the material offers enough strength while being easier to machine than higher-alloy steels.

Table 6:Pugh matrix for Connecting Rod & Upper Helping Rod

Criterion	Weight	AISI 4140	AISI 1045
Yield Strength	3	+1	+0
Machinability	1	+1	+1
Fatigue Resistance	3	+0	-1
Cost	1	+0	+1
Toughness	2	+0	-1
TOTAL		+4	-3

Down Rod & Ram:

AISI 4340 is used again for the down rod and ram to handle the full 3 MN axial force and torsion from gear drive. It resists deformation while maintaining dimensional stability under dynamic loading. Its high fatigue strength makes it suitable for repeated press cycles.

Table 7:: Pugh matrix for Down Rod & Ram

Criterion	Weight	AISI 4340	AISI 4140
Yield Strength	3	+1	+0
Fatigue Resistance	3	+1	+0
Wear Resistance	2	+1	+0
Thermal Stability	2	+1	+0
Cost	1	+0	+1
TOTAL		+10	+1

Press Base Plate:

ST52 structural steel was selected for the press base plate due to its good weldability, low cost, and sufficient strength (~500 MPa yield) to support the 3 MN distributed force without excessive deformation. It is commonly used in heavy-duty welded frames and support plates.

Table 8:Pugh Matrix for Press Base Plate

Criterion	Weight	ST52	A36
Yield Strength	3	+1	-1
Fatigue Resistance	2	+0	-1
Weldability	1	+1	+1
Machinability	1	+0	+1
Cost	1	+0	+1
TOTAL		+4	-2

6- Motor selection

The motor selection process for our link drive mechanical press was conducted using a structured approach, considering performance requirements, safety factors, integration constraints, and control capabilities. The press demands 170 kW of power and a torque of 107,150 N·m at 15 SPM, necessitating a robust drive system with a suitable gearbox. A factor of safety (FOS) of 1.2 was applied, increasing the minimum power requirement to 200–220 kW.

Table 9: Candidate Motor Specifications

Brand	Model	Power (kW)	Torque (N·m)	Speed (RPM)	Description
Siemens	1PH8 + Gearbox	250	>110,000	15	High-performance
SEW Eurodrive	DRN315M4 + KAF	200	>120,000	15	Standard industrial drive
ABB	M3BP 315SMB 4 + Gearbox	220	>110,000	15	Heavy-duty motor

Table 10:Pugh Matrix for Motor Selection

Criteria	Weight	Score (Siemens)	Score (SEW)	Score (ABB)
Power Margin	4	0	-4	-4
Torque Capability	5	0	0	-5
Control Precision	5	0	-5	-5
Reliability	4	0	-4	0

Integration & Support	3	0	3	0
Efficiency	3	0	0	0
Cost	2	0	2	2
Total Score		0	-8	-12

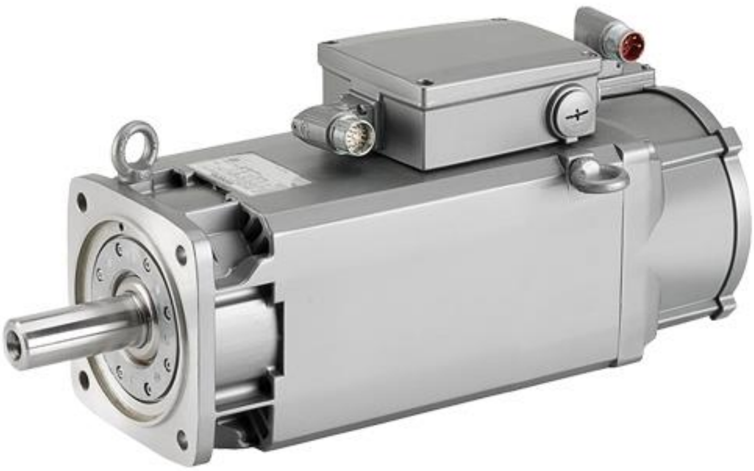

Figure 11: Siemens 1PH8 motor

Based on the evaluation criteria and Pugh matrix analysis, we can say that the Siemens 1PH8 motor system demonstrated the highest alignment with our application requirements. It offers excellent power and torque margins, advanced servo control for stroke precision, and strong support for integration with high-efficiency gearboxes. Despite its higher cost, its performance and reliability make it the optimal solution for our link drive mechanical press.

7- Manufacturing Methods

7.1. Assembly Components Bearings

Throughout the design process, we used bearings at multiple points. Our purpose was to provide bearing support for the components in the link drive section of the press machine. Below, you can see where the bearings were used and their types.

- Between Main Gear and Main Gear Shaft: **ISO 15 ABB – 38150**
- Between Eccentric Gear and Main Gear Shaft: **ISO 15 ABB – 17150**
- Between Body and Main Gear Shaft: **ISO 15 ABB – 17150**

- Between Eccentric Gear and Connecting Rod: **ISO 15 ABB – 08240**
- Between Down Rode and Connecting Rod: **ISO 15 ABB – 1780**
- Between Upper Helping Rode and Connecting Rod: **ISO 15 ABB – 1770**
- Between Upper Helping Rod and Body: **ISO 15 ABB – 1770**
- Between Down Rod and Ram: **ISO 15 ABB – 1750**

8- Static and Dynamic Calculations and Simulations

8.1 Kinematic Analysis

Kinematic analysis is a method used to examine the motion characteristics of a mechanism's components, such as position, velocity, and acceleration, over time. Through this analysis, the velocity vectors and directions of connection points (e.g., A, B, C, D) can be determined. It also provides information about how the components move, at what speed they rotate, or how far they translate linearly. Kinematic analysis is used to evaluate whether a mechanism operates efficiently and properly, and to make necessary improvements during the design process.

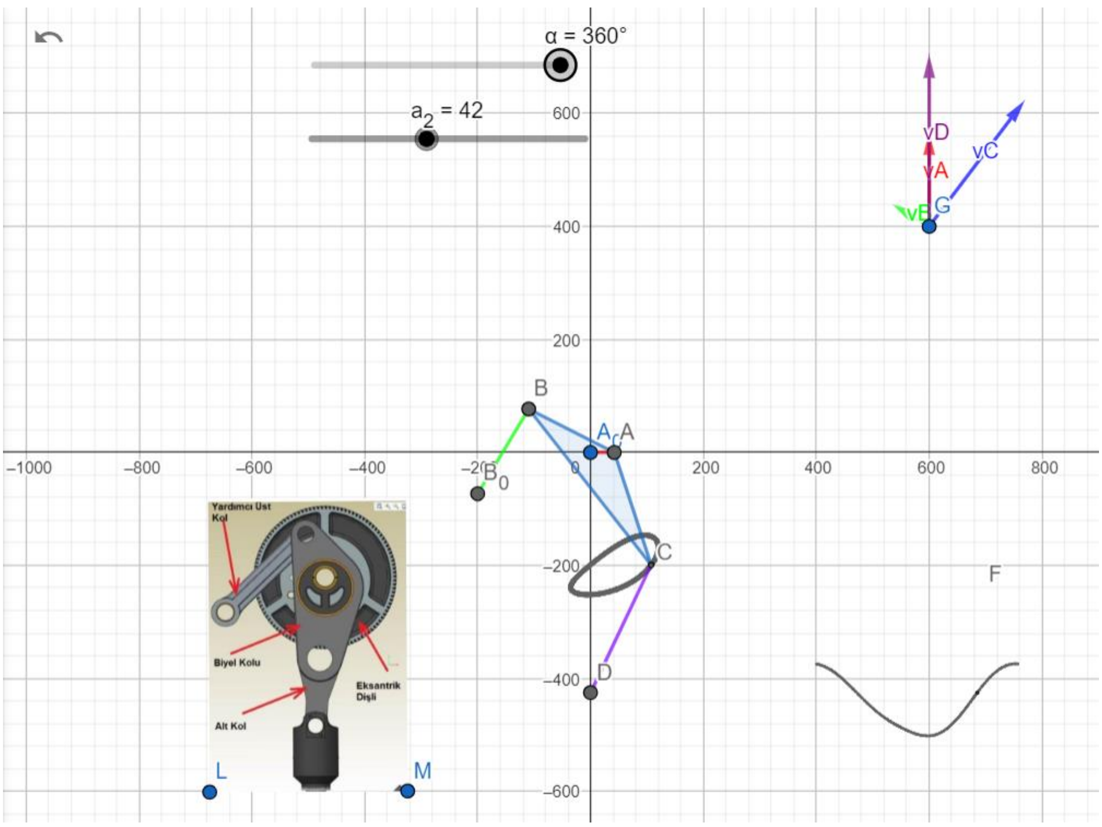

Figure 12:Analysis Picture from GeoGebra

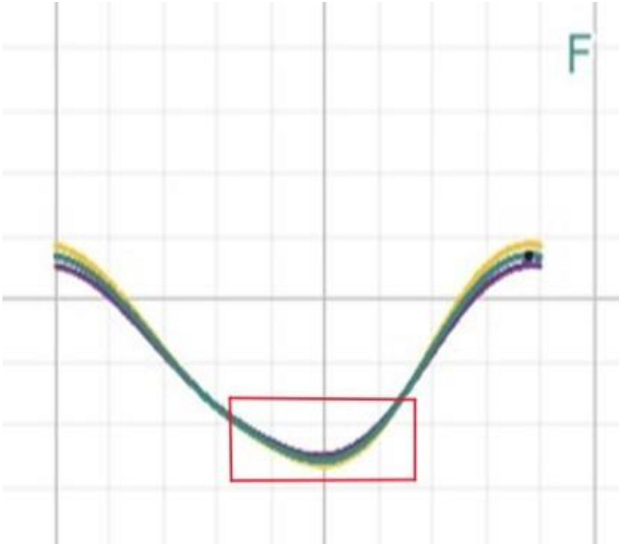

Figure 13:Stroke Profiles

These are our stroke profiles. Shown in figure (13), we can see how long we have been staying in the TDC (Top Dead Center) region. Our goal is to remain in this region for as long as possible. When we examined the three different profiles we created, it was observed that the green profile spends more time in the TDC region. Therefore, we determined that the green one is the most optimal profile.

Stroke profile:

The stroke profile defines the up-and-down motion of the ram. An accurate stroke function ensures smooth and predictable press operation, minimizing vibrations and ensuring consistent force application during forming.

8.1.1 Kinematic Analysis (Stroke)

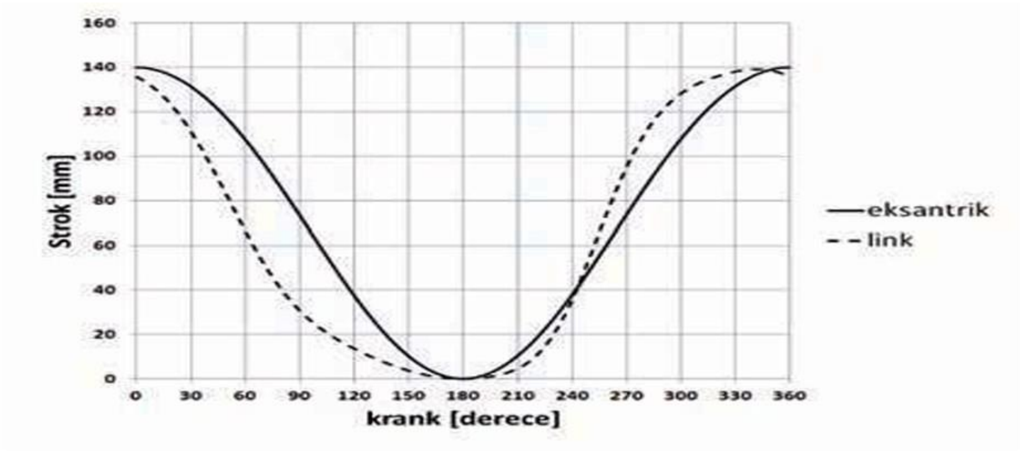

Figure 14:Kinematic Analysis stroke graph for link drive mechanism

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The graph represents the curve of the stroke profile of the Link Drive Mechanical press in terms of Theta. So, if we look at each angle in the X-axis we can see its corresponding stroke displacement in the Y-axis. On the other hand, this curve gives us S (θ).

Table 11:Kinematic Analysis stroke (Angle vs stroke)

Angle (θ).	Stroke (mm)
0	140
30	110
60	62.5
90	33
120	15
150	4
180	0
210	5
240	35
270	95
300	108
330	138
360	140

Using the table above, we can estimate the values (x, y) if the max stroke is 130.

Substituting the stroke values into this equation: New_Stroke = Stroke $\times$ (130 / 140).

we get table (12) below stating our new stroke values:

Table 12: New stroke values calculated by MATLAB

Angle ().	Stroke (mm)
0	130
30	102
60	58.04
90	30.86
120	13.93
150	3.71
180	0
210	4.64
240	32.5
270	88.39
300	100.71
330	128.57
360	130

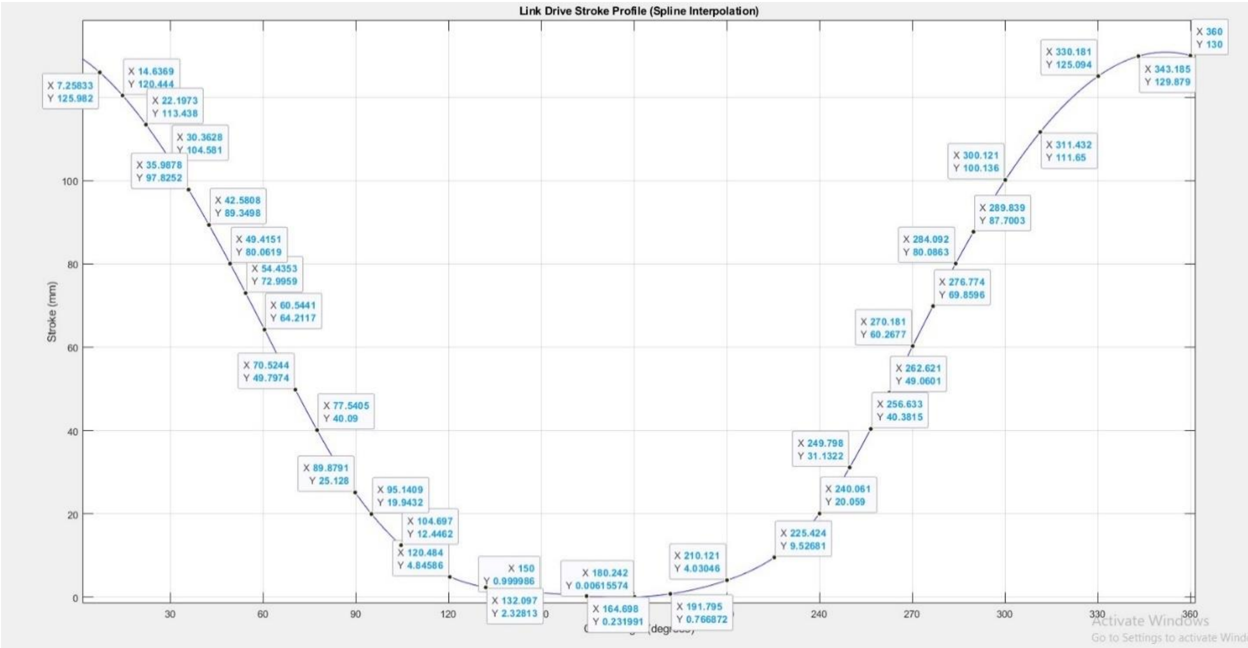


Figure 15:Link drive stroke profile (Spline interpolation)

8.1.2 Kinematic Analysis (Velocity)

The curve shown in figure (15) represents the stroke function. To make it more accurate, a huge number of random coordinates were chosen as shown in the figure above.

By using MATLAB software, it gave us a polynomial form in the 8th order:

[0.0000 -0.0000 0.0000 -0.0000 0.0000 -0.0006 0.0017 -0.5657 129.8591]

$$S(\theta) = -0.006X^3 \quad + \quad 0.0017X^2 \quad - \quad 0.5657X \quad + 129.8591$$

It also gives us the most accurate curve as shown below in figure (16).

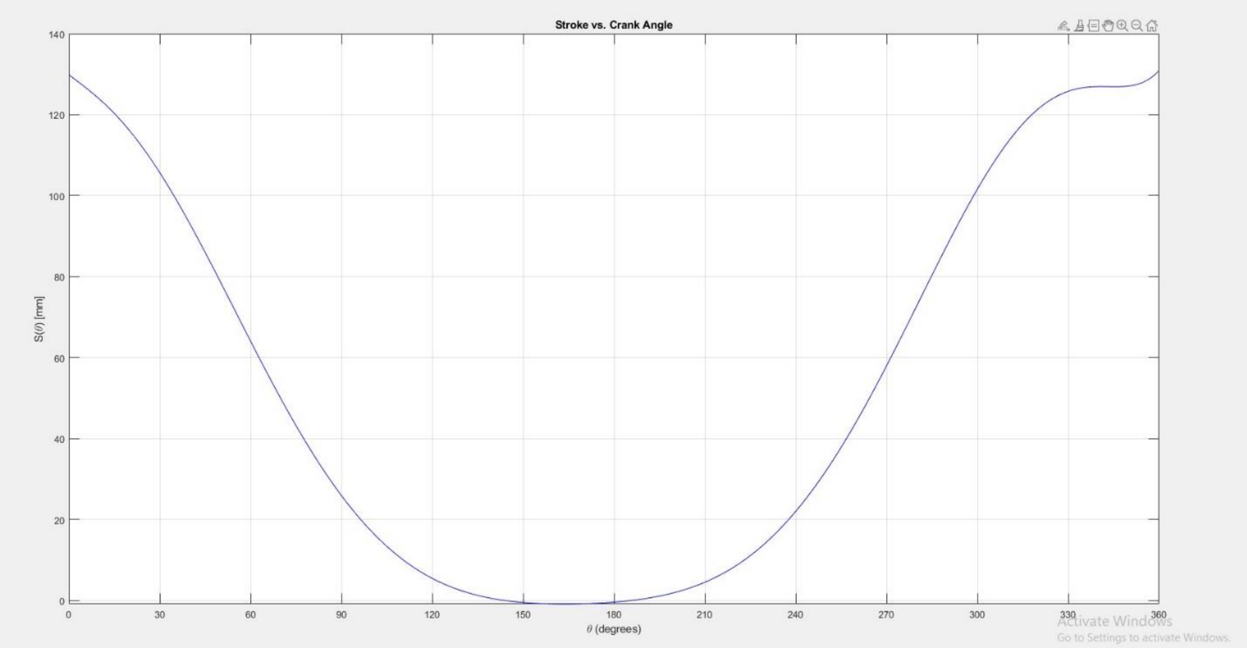


Figure 16:Stroke vs. Crank angle graph

8.1.3 Kinematic Analysis (Acceleration)

Now since we have the displacement equation, we can easily obtain the velocity equation by differentiating S (θ).

$$V(\theta) = -0.0018X^2 + 0.0034X - 0.5657$$

Now, substituting the same angle inputs in the previous tables into the velocity equation we get the below curve, shown in figure (17):

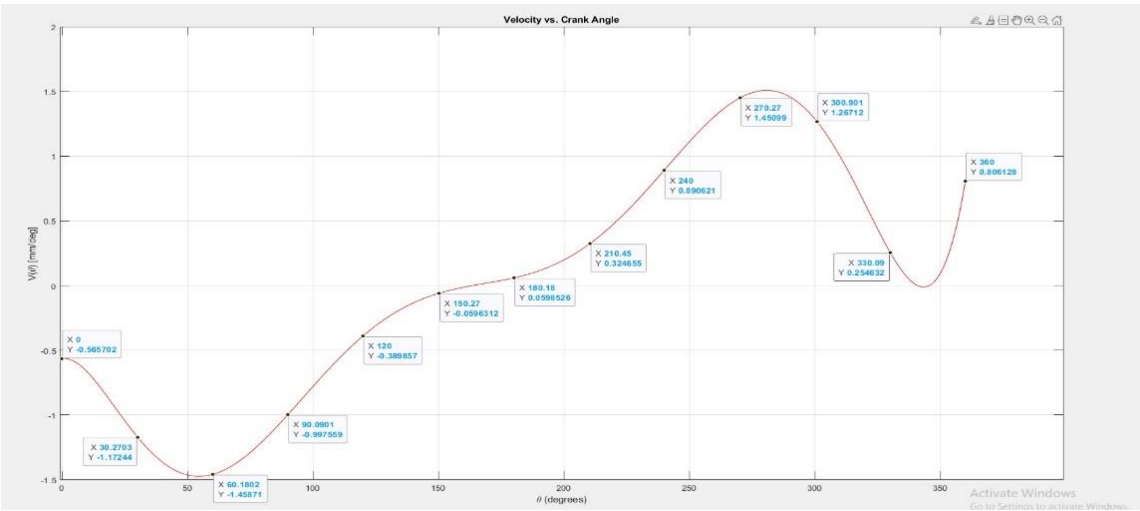


Figure 17:Velocity vs. crank angle graph

Now, since we have the Velocity equation, we can obtain acceleration by differentiating V (θ)

$$\text{Acceleration } (\theta) = -0.0036X \quad + 0.0034$$

Substituting the same angle inputs in the previous tables into the velocity equation we get the curve below as figure (18) shows:

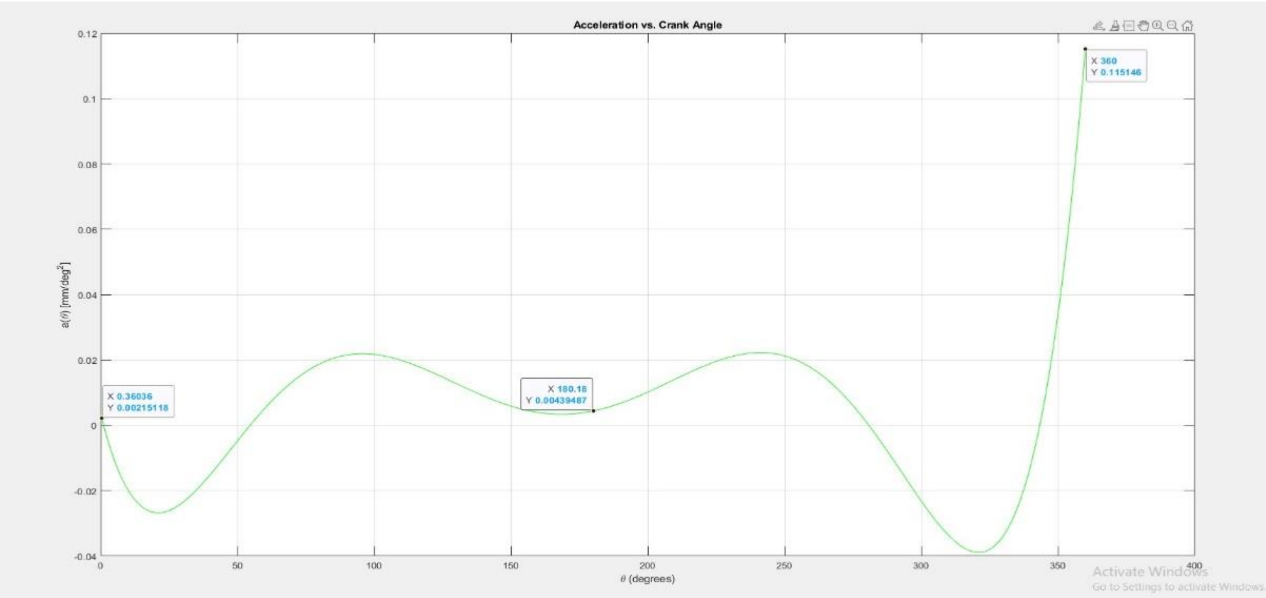


Figure 18:Acceleration vs. Crank angle graph

8.2 Stress and Safety Analysis

8.2.1 Main Gear stress analyses

We mentioned before in the constraints part that stroke = 140 mm (The maximum Vertical (Up-Down) distance of the ram), Stroke per min (N) = 15 spm, maximum force to transferee 3000 KN. Torque and power calculations help confirm that the gear can safely transmit the required energy. They also guide the selection of gear material and dimensions to avoid failure under load. The force per tooth and number of teeth in contact determine how load is distributed, affecting wear and safety margins.

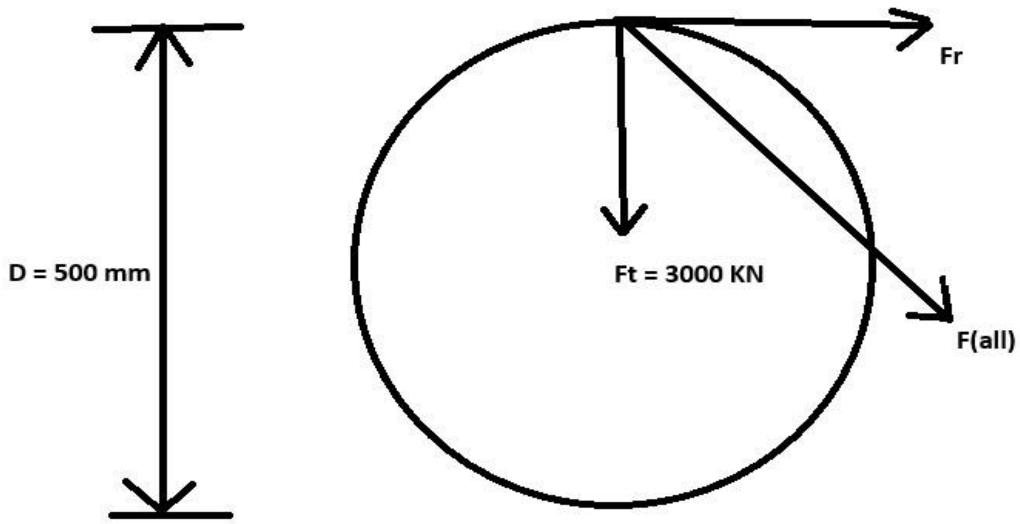


Figure 19: Free diagram of main gear

Angular velocity (w) is calculated to relate stroke rate to rotational speed:

$$w = \frac{2\pi \times N}{60} = \frac{2\pi \times 15}{60} = 1.57 \text{ rad/s},$$

where N = 15 spm

Angular velocity links how fast a mechanism, like a press or piston, moves back and forth to the speed at which the gear turns. It keeps the back-and-forth motion and the turning motion in sync.

Since the gear force is distributed among several teeth in contact, the effective number of teeth (Zc) in contact is

$$Z_c = 1.5 \times \sqrt{Z} = 1.5 \times \sqrt{20} \approx 7$$

In spur gears, several teeth usually engage at the same time. This helps spread out the load and lowers stress on any single tooth. The Zc value gives an idea of how many teeth are sharing the load, which is key for figuring out contact stresses and how long the gear will last.

The force transmitted per tooth **(F_t)** becomes:

$$F_t = \frac{F}{Z_c} = \frac{3\times10^6}{7} = 428.6\ kN \text{ per teeth}$$

This is the usual load each tooth deals with while meshing. It's crucial to make sure each tooth can handle this force to avoid gear failure from too much stress.

The torque (T) transmitted by the gear is:

$$F_t \times r = 428.6 \times 10^3 \times 0.25 = 107150\ N.m$$

Torque is how much rotational work a gear can do. It's key for figuring out the size of the shaft, the material for the gear, and how strong it needs to be to work properly.

The power (P) delivered is:

$$P = T \times w = 107150 \times 1.57 = 168225.5 \text{ W} = 168.23 \text{ KW}$$

ω=1.57rad/s (angular velocity)

This is how fast work gets done or energy moves in a rotating system. It's important for designing motors and gears to make sure they provide enough power for what they're used for. To make sure the gear shaft can handle the torque, we figure out the torsional shear stress (τ) on the shaft using the torsion formula:

$$\tau = \frac{T}{J} \times r$$

Where:

1) T = Torque transmitted by the gear.

2) r = gear pitch radius.

3) J = Polar moment of inertia. = $J = \frac{\pi \times D^4}{32}$

$$\tau = \frac{107150}{\frac{\pi \times 0.25^4}{32}} \times 0.25 = 70\ MPa$$

This value should be compared with the shear yield strength of the shaft material. If τ is significantly lower than the yield limit (with an appropriate factor of safety), the design is considered safe under torsional loading. Otherwise, an increase in shaft diameter or material strength would be necessary. To figure out how strong gear teeth are when bending, people usually use the Lewis Equation. This equation helps calculate the highest bending stress at the root of the gear tooth caused by the load. Here's the bending stress formula mentioned below:

$$\sigma_{bending} = \frac{W_t}{Y \times b \times m}$$

Where:

b = 150 mm (face width)

m = 25

W_t = 428.5 kN

Y = 0.31 (Lewis's form factor for Z=20)

$$\sigma_{bending} = \frac{F_t}{Y \times b \times m} = \frac{428600}{0.31 \times 150 \times 25} = 366 \text{ MPa}$$

Increasing the face width can further reduce the bending stress, enhancing the gear's durability. The bending stress at the root of the gear tooth is around 366–369 MPa. You need to check this against the bending strength of the gear material. To make sure it lasts, the yield strength of the material should be a good bit higher than this stress, especially when factoring in safety.

8.2.2. Main gear total deformation and modal analysis

The finite element analysis (FEA) conducted on the main gear was essential to assess whether it could safely transmit the 3000 kN pressing force and rotational torque without exceeding allowable stress limits. As a core transmission element, the main gear is subjected to high tangential forces during meshing, with additional stress concentration at the tooth roots and bore center, as shown in figure (20) below.

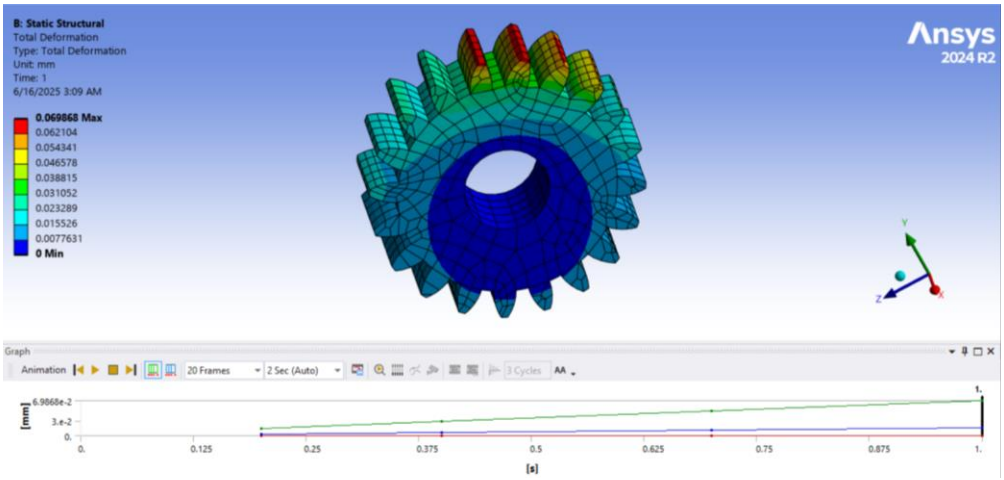


Figure 20:: Total deformation analysis for the main gear

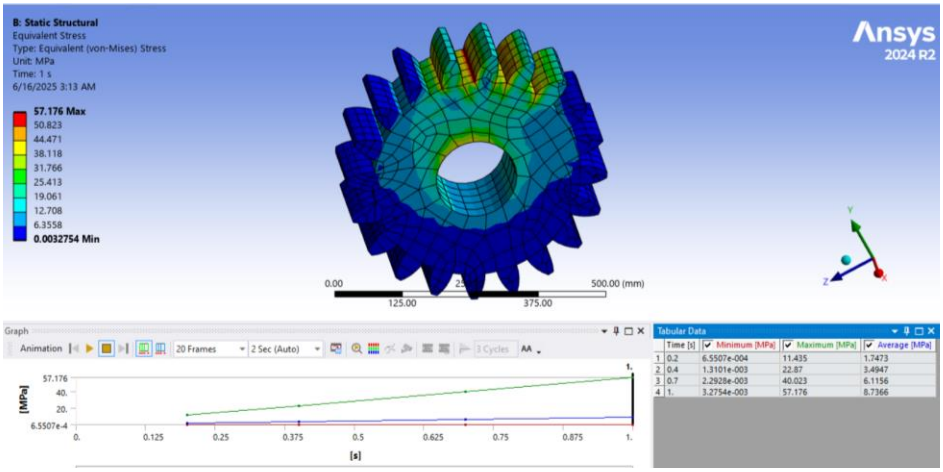


Figure 21:Equivalent (Von-Mises) stress analysis for main gear

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Main Gear	Static	428,600 N tangential force, torque from shaft	Fixed at bore (shaft mount)	AISI 4340

Figure 22:Analysis proprieties for the main gear

From the ANSYS static structural analysis, the total deformation of the main gear under full load was minimal, validating its geometric and material adequacy we selected. The Von Mises stress distribution as shown in figure (21), revealed stress peaks localized around the gear teeth and bore regions, yet all values remained well within the yield strength of AISI 4340

(≈850 MPa). This confirms the gear's mechanical robustness, in line with hand calculations predicting a bending stress of ~366 MPa and total stress around 386 MPa, which was used to calculate a FoS of 1.69.

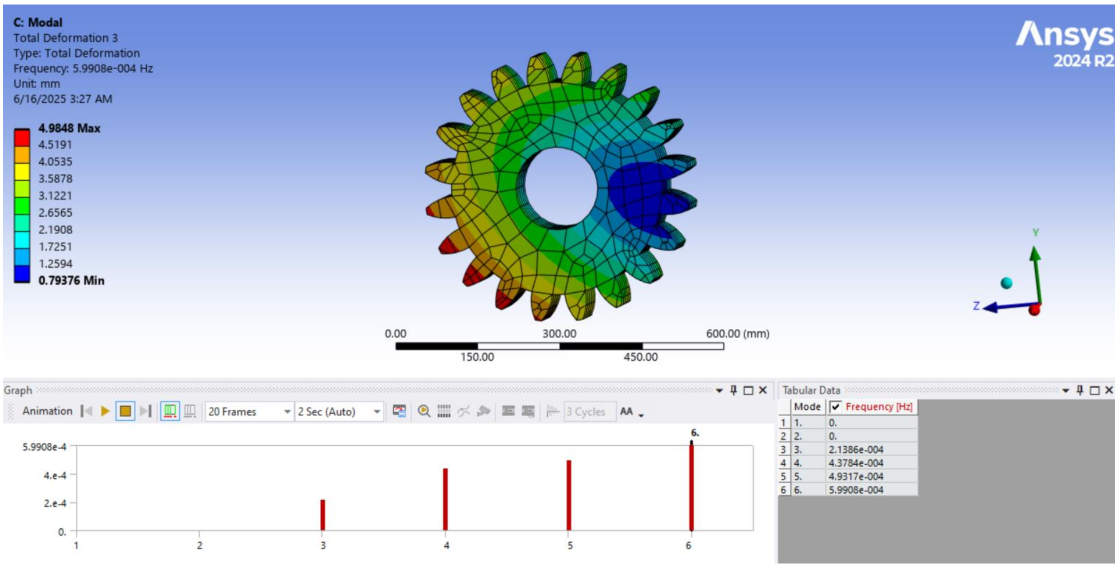


Figure 23:Modal analysis for main gear

The modal analysis of the main gear showed its 6th natural frequency at approximately 5.99e-4 Hz, which is far above the operational excitation frequency of the system (0.25 Hz from 15 SPM). This confirms no resonance condition will occur during operation. Thus, the FEA simulation validates both the static strength and the dynamic safety of the gear under industrial working conditions.

8.2.3. Combined Stress Analysis (Von Mises Criterion)

To figure out the overall impact of bending and twisting stresses, we calculate the equivalent stress using the Von Mises formula:

$$\sigma_{all} = \sqrt{\sigma_{bending}^2 + 3 \times \tau^2} = 386\,MPa$$

8.2.4 Main Gear Safety Analysis

The total stress on the gear tooth is about 386 MPa. This number is important because we need to compare it with the strength of the gear material to make sure it works safely and reliably.

Static Factor of Safety (FoS)

To ensure that the selected material can withstand the applied stress under static loading conditions, the static factor of safety is calculated:

$$FOS = \frac{\sigma_{yield}}{\sigma_{applied}} = \frac{850}{386} = 2.2$$

Maximum Force = 3 MN, Minimum Force = 2 MN

Maximum Stress = 386 MPa, Min Stress = $\frac{2}{3} \times 386 = 258\,MPa$

Endurance Limit Se = 300MPa, Yield Strength Sy =850MPa

We can use Soderberg Criterion to find the factor of safety:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{Sy} = \frac{1}{FOS}$$

$$\frac{64}{300} + \frac{322}{850} = \frac{1}{FOS} = \text{FOS} = 1.69$$

The Soderberg criterion is a conservative method for fatigue analysis, combining mean and alternating stresses. A Factor of Safety (FOS) of 1.69 indicates a safe design, especially in industrial machinery where unexpected overloads might occur.

8.3 Main Gear Shaft Stress Analysis

The shaft connected to the main gear is subjected to two primary stresses:

- Bending stress caused by the transmitted force from the gear teeth.
- Torsional stress resulting from the torque applied to the shaft.

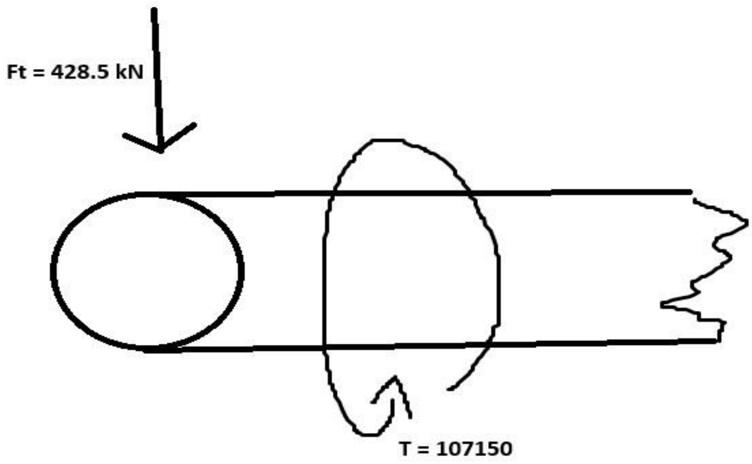


Figure 24:Free body diagram of main gear's shaft

Shafts experience both bending (due to force from the gear) and torsion (due to transmitted torque). The Von Mises stress formula combines these to evaluate failure risk under complex loading. This approach is essential for parts subjected to multiaxial stress states. The bending stress on a circular shaft is given by:

$$\sigma_{bending} = \frac{M}{Z} = \frac{F_t \times L \times 32}{\pi D^3} = 646 \text{ MPa}$$

Where:

Ft = 428600 kN

L = 0.5 m

D = 0.15 m

$Z = \frac{C}{I} = \frac{D/2}{\pi \times D^4/64} = \frac{32}{\pi \times D^3}$

$$\sigma_{bending} = \frac{M}{Z} = \frac{F_t \times L \times 32}{\pi D^3} = \frac{428600 \times 0.5 \times 32}{\pi \times 0.15^3} = 646 \text{ MPa}$$

The torsional (shear) stress on the shaft is given by:

$$\tau = \frac{T}{W} = \frac{T \times 16}{\pi \times D^3}$$

Where:

T = the torque = 107150 (Found above)

$$W = \frac{y}{J} = \frac{\frac{D}{2}}{\frac{\pi \times D^4}{32}} = \frac{16}{\pi \times D^3} \qquad \text{where, } D = 0.15 \text{ m}$$

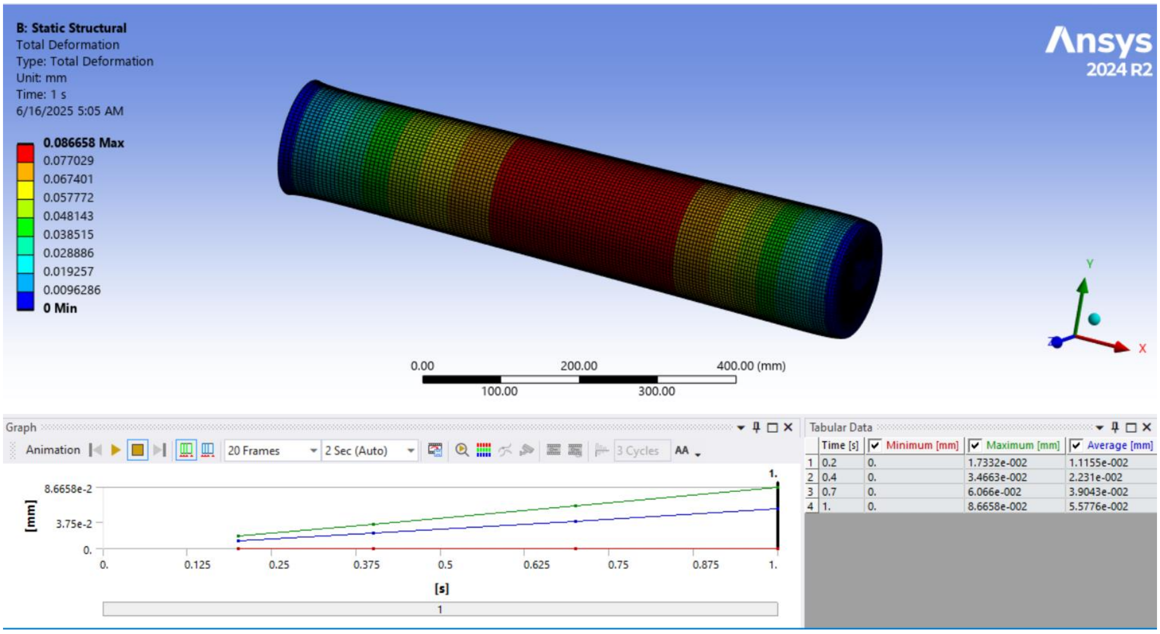
$$\tau = \frac{T}{W} = \frac{T \times 16}{\pi \times D^3} = \frac{107150 \times 16}{\pi \times D^3} = 162 \text{ MPa}$$

To account for both bending and torsional stresses, the equivalent (von Mises) stress is calculated:

$$\sigma_{all} = \sqrt{\sigma_{bending}^2 + 3\tau^2} = 705 \text{ MPa}$$

$$= \sqrt{\left(\frac{428600 \times 0.5 \times 32}{\pi \times 0.15^3}\right)^2 + 3 \times \left(\frac{107150 \times 16}{\pi \times 0.15^3}\right)^2} = 705 \text{ MPa}$$

8.3.1. Main haft total deformation



	Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1	0.2	0.	1.7332e-002	1.1155e-002
2	0.4	0.	3.4663e-002	2.231e-002
3	0.7	0.	6.066e-002	3.9043e-002
4	1.	0.	8.6658e-002	5.5776e-002

Figure 25:Total deformation analysis for the main gear’s shaft

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Main Gear Shaft	Static	3 MN Axial, 107,150 Nm Torque, 420,000 Nm Bending	Fixed at both ends	Maraging steel 300

Figure 26: Analysis proprieties for the main gear's shaft

8.3.2 Main Shaft Safety Analysis

The shaft is under a combined stress of 705 MPa. We need to check this against the material's yield strength while applying a safety factor to make sure it works well in real-life situations. Maraging Steel 300 is known for its exceptional strength, toughness, and fatigue resistance, making it suitable for high-performance mechanical components such as shafts in heavy-duty applications.

- Sy =1860MPa
- Sut =2068MPa
- Se ≈900MPa

Static Factor of Safety (FoS)

$$FOS = \frac{\sigma_{yield}}{\sigma_{applied}} = \frac{1850}{705} = 2.62$$

Fatigue Safety Check Using Modified Goodman/Soderberg Criterion:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FOS}$$

$$\frac{149.17}{900} + \frac{745.83}{2068} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 2.24$$

The shaft maintains a fatigue factor of safety of 2.24, ensuring safe long-term operation under cyclic loading. This is considered very reliable in high-performance engineering systems. Shaft dimensions are selected to minimize stress while balancing weight and cost. A larger diameter reduces stress but increases mass. The chosen dimensions aim for optimal performance under the calculated load while ensuring durability and fatigue resistance. The current design works well mechanically, handling both static loads and fatigue while keeping the structure strong for its entire lifespan.

8.4 Pinion Stress Analysis

The pinion meshes with the main gear. Therefore, the modulus of it is fixed to match the analyses done before where $m_p = 2.5\,mm$ where the tooth profile is safely fixed to be 100 teeth for pinion with standard involute.

As it is calculated before, the tangential force acting on the gear is $W_t = 428.6\,KN$

And as is calculated before, the Torque (T) need to be transmitted is T = 107150 N.m

Tortional stress analyses:

$$\tau = \frac{T}{J} \times r$$

1) T = Torque transmitted by the gear.

2) r = gear pitch radius.

3) J = Polar moment of inertia. $J = \frac{\pi \times D^4}{32}$

$$\tau = \frac{107150}{\frac{\pi \times 0.2}{32}} \times 0.2 = 137\,MPa$$

The most affective stress acting on the pinion is Torsional stress.

$$FOS = \frac{\sigma_{yield}}{\sigma_{applied}} = \frac{1900}{558} = 13$$

Using Soderberg formula to compare the factor of safety:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{FOS}$$

$\sigma_m = \frac{137+91}{2} = 114$ .... $\sigma_a = \frac{137-91}{2} = 23$

$$\frac{91}{800} + \frac{137}{1900} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 5.4$$

8.5 Eccentric Gear Stress Analysis

The stresses acting on the eccentric gear are analyzed. Due to the offset (eccentricity) between the shaft center and the gear center, both bending, torsional, and compressive stresses occur.

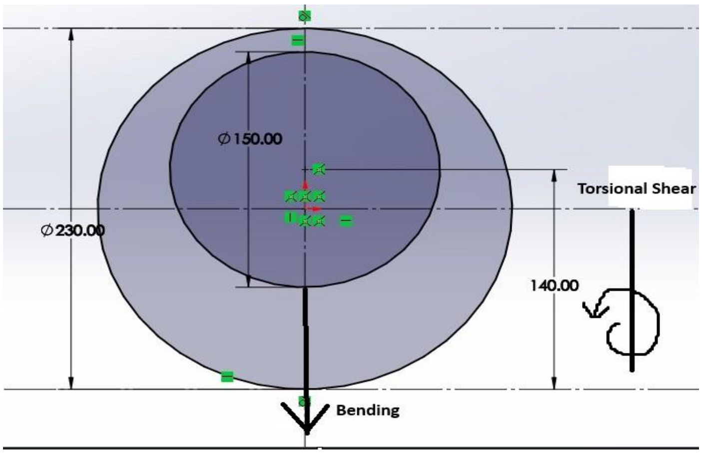


Figure 27:Eccentric gear torsional sheer diagram

In systems with eccentric gears, the center of the gear doesn't line up with the center of the shaft. This offset creates a bending moment on top of the twisting force that's already acting on the shaft. When force is applied at the center of the gear, the shaft still spins around its own axis. This causes a distance between where the force is applied and the center of the shaft, leading to that bending moment.

8.5.1. Bending Moment and Bending Stress Calculation:

The bending moment *(M)*is a measure of the internal moment that causes a structural element to bend. It is calculated using the equation:

$$M = F \times e$$

Where.

F = Applied force = 3000 kN

e = 140 mm = Eccentricity (the perpendicular distance from the line of action of the force to the neutral axis)

$$M = F \times e = 3 \times 10^6 \times 140 = 420000 \text{ N.m}$$

Height (h): 90 mm, thickness (t): 300 mm, eccentricity (e): 140 mm, Force (F): 3 MN

Thus, the bending stress is:

$$\text{Bending stress } \sigma_{bending} = \frac{M}{Z} = \frac{F \times e}{Z}$$

Where $Z = \frac{I}{y}$, and (e) is the gear eccentricity

First, the moment of inertia (I) is found for the rectangular section:

$$(I) = \frac{t \times h^3}{12} = \frac{200 \times 140^3}{12} = 45.73 \times 10^6 \, mm^4$$

Then, the distance to the neutral axis: $y = \frac{h}{2} = \frac{140}{2} = 70 \, mm$

So, the section modulus:

$$Z = \frac{I}{y} = \frac{45.73 \times 10^6}{70} = 653000 \, mm^3$$

Now, the bending stress:

$$\sigma_{bending} = \frac{M}{Z} = \frac{F \times e}{Z} = \frac{3 \times 10^6 \times 140mm}{653000} \approx 643 \, MPa$$

The maximum bending stress from a 3000 kN force applied 140 mm from the neutral axis is about 642.83 MPa. You should check this against the yield strength of the material to make sure the design is safe.

Torsional stress τ arises when a torque is applied to a shaft or structural member. It is calculated using the formula:

$$(\tau) = \frac{T \times c}{J}$$

Where:

1) T is the torque, and it was found before = 107150 N.m
2) c = $\frac{h}{2} = \frac{140}{2} = 70 \, mm$ = Distance from the center to the outermost fibe
3) The polar moment of inertia (J) for a rectangle is: =

$$\frac{th^3}{3}\left[1 - 0.63\frac{h}{t}\right] = \frac{200 \times 140^3}{3}\left[1 - 0.63\left(\frac{140}{200}\right)\right] = 102.3 \times 10^6\, mm^4$$

Torsional stress:

$$(\tau) = \frac{T \times c}{J} = \frac{107150 \times 10^3 \times 45}{102.3 \times 10^6} = 105\, MPa$$

$$(\tau) = \frac{T \times c}{J} = \frac{107150 \times 10^3 \times 45}{21.3 \times 10^6} = 226\, MPa$$

Direct compressive stress occurs due to vertical axial loading:

$$\sigma_{compressive} = \frac{F}{A}$$

Where:

1) F = 3 kN
2) (A) is the cross-sectional area

$$\text{A} = t \times h = 200 \times 140 = 28000 mm^2$$

$$\sigma_{compressive} = \frac{F}{A} = \frac{3 \times 10^6}{28000} = 107\, MPa$$

The total normal stress acting on the gear is the sum of bending and compressive stresses:

$$\sigma_{total} = \sigma_{bending} + \sigma_{compressive}$$

$$\sigma_{total} = 643 + 107 = 750\, MPa$$

Finally, to combine total normal and torsional stress (shear), the equivalent von Mises stress is calculated:

$$\sigma_{all} = \sqrt{\sigma_{total}^2 + 3\tau^2} = \sqrt{750^2 + 3 \times 105^2} = 770\, MPa$$

8.5.2. Eccentric gear total deformation

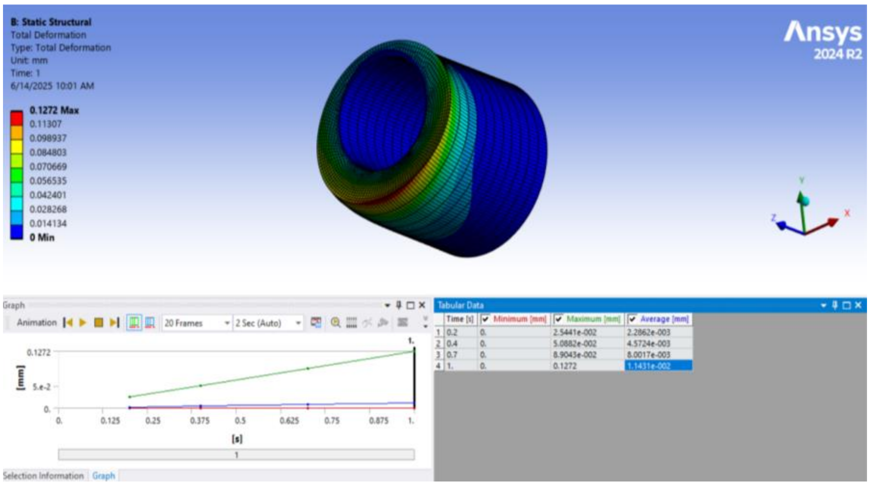


Figure 28:Total deformation analysis for the eccentric gear.

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Eccentric Gear	Static	3 MN axial, 420,000 Nm moment	Fixed at bore	AISI 4340

Figure 29:Analysis proprieties for the eccentric gear

Static deformation analysis in figure (28) above shows small elastic deflections, which indicates structural stability. The equivalent stress distribution was highest around the eccentric bore region, which aligns with expected stress concentrations due to eccentric loading. The FEA-calculated stresses matched the manual bending stress estimates (~642 MPa) and remained within the AISI 4340 yield strength limit (740–850 MPa), supporting the design's safety.

8.5.3 Eccentric Gear Safety Analysis

The selected Material is AISI 4340, thus:

$$FOS = \frac{\sigma_{yield}}{\sigma_{applied}} = \frac{1900}{705} = 2.69$$

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FOS}$$

$$\frac{117.5}{800} + \frac{587.5}{1900} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 2.19$$

8.6 Connecting Rod Stress Analysis

The connecting rod is the link between the eccentric gear and the down-moving rod. It experiences a combination of bending, axial, and shear stresses due to the eccentric forces transmitted through the system.

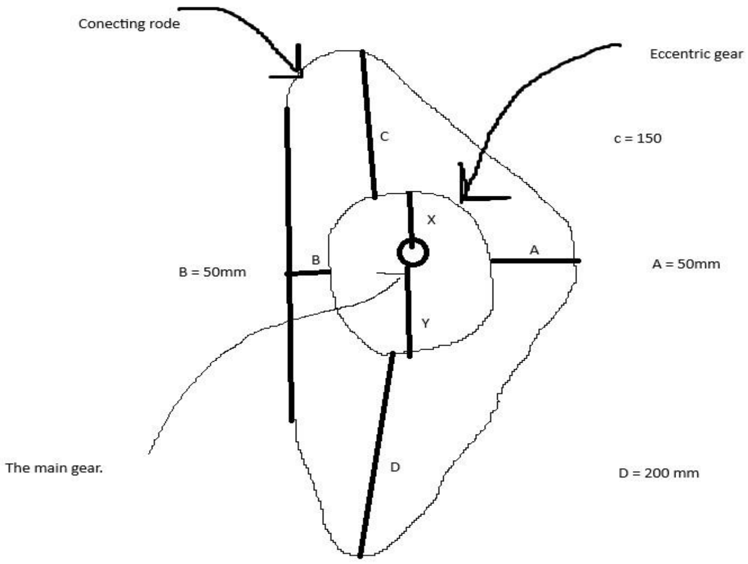


Figure 30:Connecting rod analyzing diagram

Point D was selected for analysis because it is farthest from the center and experiences the maximum bending moment. Understanding the force distribution across the rod ensures the design remains safe under all loading positions.

A = 50mm = the distance from the bottom of the gear to the bottom of the rode

B = 50mm = the distance from the bottom of the gear to the bottom of the rode

C = 150mm = the distance from the bottom of the gear to the bottom of the rode in +Y-axis

D = 200mm = the distance from the bottom of the gear to the bottom of the rode in -Y-axis

X=100mm= distance from the center of the shaft to the upper bottom of the rode \+Y-axis

Y = 100mm= the distance from the center of the shaft to the upper bottom of the rode in –Y-axis

The maximum force is expected to act at point D (at the farthest end along -Y axis), where the bending moment and axial load are highest. Other points (A, B, C) will experience lower stresses; thus, the design is mainly based on point D.

The following types of stress are considered:

a) Bending Stress Bending stress occurs due to the force applied at a distance from the neutral axis.

The bending moment is:

$$M = F \times D$$

The bending stress formula for a rectangular section is:

$$\sigma_{bending} = \frac{M \times C}{I} = \frac{F \times D \times C}{\frac{1}{12} b t^2}$$

- C = t/2 = distance from neutral axis to surface
- I = $\frac{1}{12} b t^3$ = moment of inertia for a rectangle
- t= 200 mm (thickness)
- b = 200 mm (width

$$\sigma_{bending} = \frac{M \times C}{I} = \frac{F \times D \times C}{\frac{1}{12} b t^3} = \frac{3 \times 10^6 \times 200 \times (200/2)}{\frac{1}{12} \times 200 \times 200^3}$$

$$= 450\, MPa$$

We calculated the bending stress because the force applied at **D** causes a bending moment around the neutral axis.

Axial stress occurs:

directly from the transmitted vertical force. The formula:

$$\sigma_{axial\,shear} = \frac{F}{A}$$

Where: t = the thickness = 150mm

b = the width = 200 mm

F = the force = 3MN

$$\sigma_{axial\,shear} = \frac{F}{A} = \frac{F}{t \times b} = \frac{6 \times 10^6}{200 \times 150} = 100MPa$$

Since bending stress and axial stress are both normal stresses and act in the same direction, they are added directly:

$$\sigma_{all} = \sigma_{axial} + \sigma_{bending} = 550MPa$$

8.6.1. Connecting rod total deformation

The connecting rod experiences complex stress states as it links the eccentric motion to the downward ram movement. Bending and axial compression dominate due to the transmission of the 3 MN force.

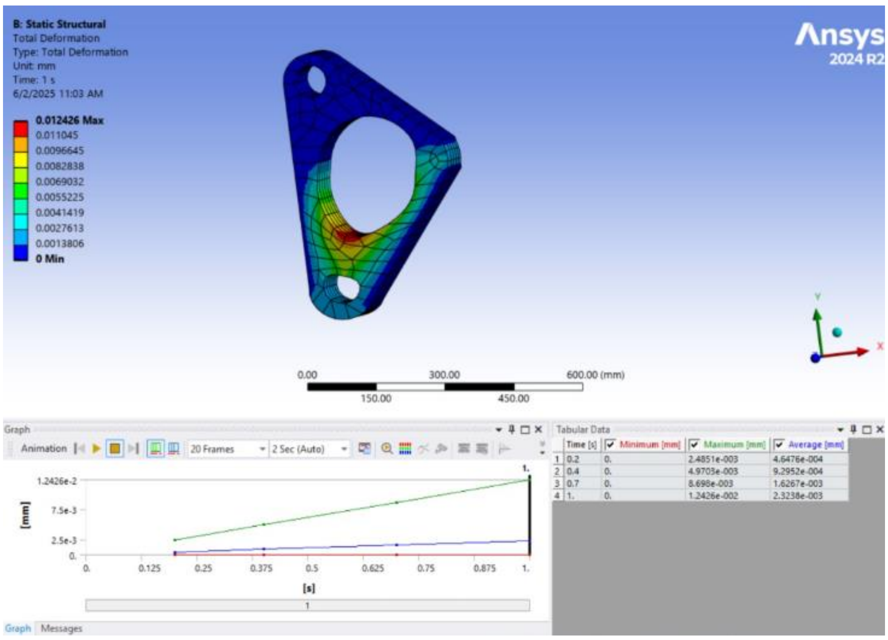


Figure 31:: Total deformation of the connecting rod

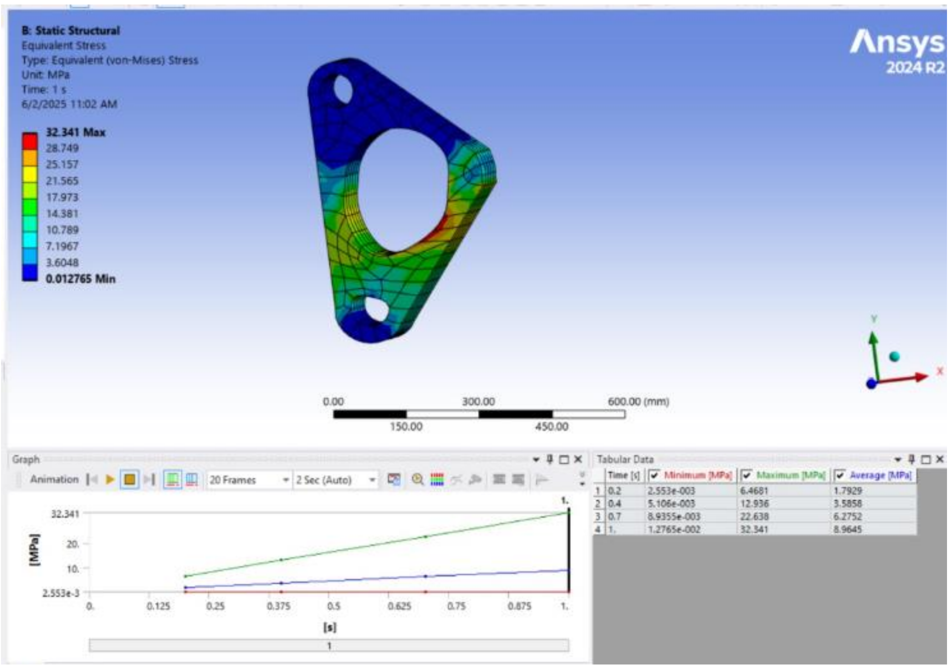


Figure 32:: Equivalent (Von-Mises) stress analysis for connecting rod

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Connecting Rod	Static	3 MN Axial, Bending at joint D	Pinned ends	AISI 4140

Figure 33:Analysis proprieties for the connecting rod

The deformation shown in figure (30), indicated moderate deflection in line with rod flexibility, while the Von Mises stress was concentrated near the mounting flanges and shaft interface. Compared to the calculated maximum combined stress (~824 MPa), the simulated stress stayed within the yield strength of AISI 4140 (860 MPa), ensuring acceptable safety under peak loads.

8.6.2 Connecting Rod Safety Analysis

- Yield Strength Sy ≈ 860 MPa
- Ultimate Strength Sut ≈ 1000MPa
- Endurance Limit Se = 500 MPa

$$FOS = \frac{\sigma_{yield}}{\sigma_{applied}} = \frac{860}{550} = 1.56$$

$$\sigma_m = \frac{550+366.67}{2} = 485.33 \quad ,,, \quad \sigma_m = \frac{550-366.67}{2} = 91.67$$

Appling Goodman Criterion to find the FOS:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FOS}$$

$$\frac{91.67}{500} + \frac{458.33}{860} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 1.39$$

8.7 Upper Helping Rod Stress Analysis

Even though the helping rod is not a primary load-bearing component, it still experiences some axial and bending stresses due to reactions during operation.

The axial stress:

$$\sigma_{axial} = \frac{F}{A} = \frac{F}{b\,.\,t}$$

$$\sigma_{axial} = \frac{F}{A} = \frac{F}{b\,.\,t} = \frac{600000}{100 \times 70} = 85.71\,MPa$$

- F = 60000 (estimated reaction force on the rod)
- A=b×t (cross-sectional area)
- b=100 mm (width)
- t=70 mm (thickness)

$$\sigma_{axial} = \frac{F}{A} = \frac{F}{b\,.\,t} = \frac{600000}{100 \times 70} = 85.71\,MPa$$

The Bending stress occurs due to external forces acting with an offset.

Bending stress is calculated by:

$$\sigma_{bending} = \frac{M}{Z} = \frac{\frac{(F \times L)}{4}}{\frac{b.t^2}{6}}$$

$$\sigma_{bending} = \frac{M}{Z} = \frac{(F \times L)}{\frac{b.t^2}{6}} = \frac{60000 \times 400}{\frac{100 \times 70^2}{6}} = 294\ MPa$$

Since axial and bending stresses are both normal stresses:

$$\sigma_{EQV} = \sigma_{axial} + \sigma_{bending} = 380\ MPa$$

8.7.1. Upper helping rod total deformation analysis

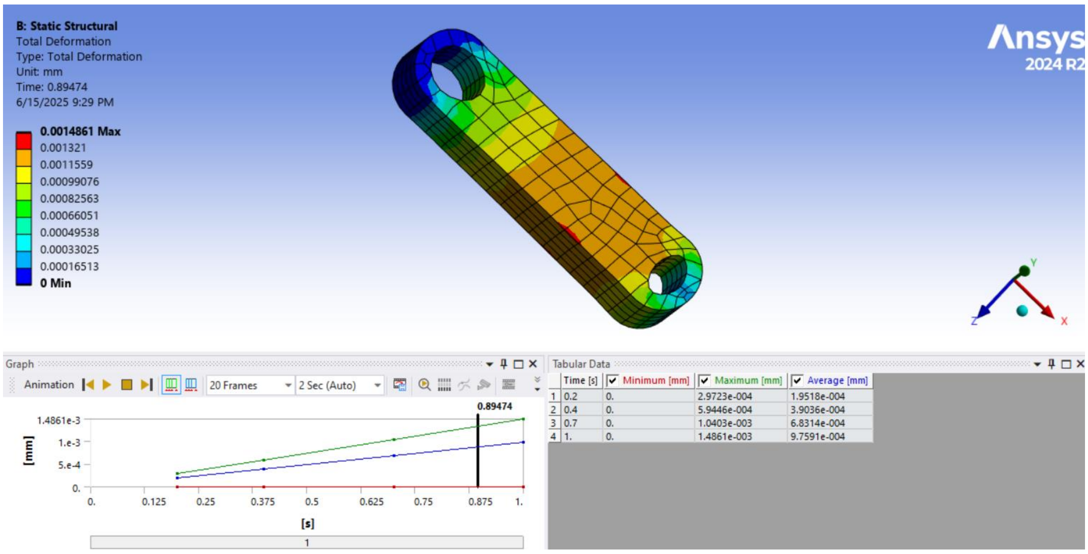


Figure 34:: Total deformation of the upper helping rod

Analyzing figure (34) above, we can check that the total deformation was minor, and the stress distribution showed localized bending zones near fixed connections. All values were under the 860 MPa yield strength of AISI 4140, confirming safety and functionality without overengineering.

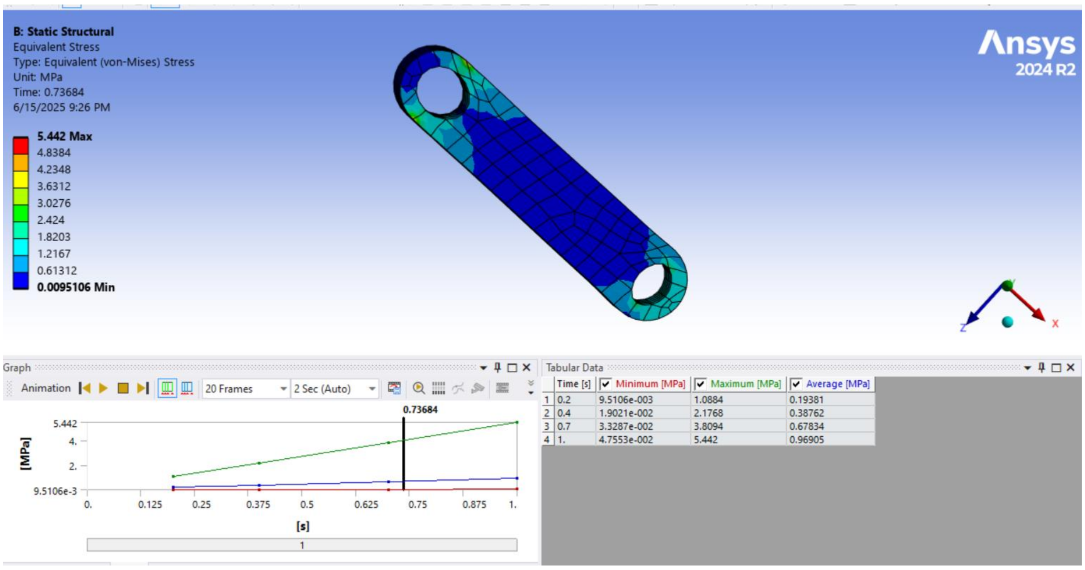


Figure 35:: Equivalent (Von-Mises) stress analysis for upper helping rod

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Upper Helping Rod	Static	60,000 N axial	Pinned ends	AISI 4140

Figure 36:Analysis proprieties for the upper helping rod

8.7.2 Upper Helping Rod Safety Analysis

- Yield Strength Sy ≈860MPa, Ultimate Strength Sut ≈1000 MPa
- Endurance Limit Se ≈500MPa

$$FOS = \frac{Sy}{\sigma} = \frac{860}{380} = 2.26$$

Now using Goodman Criterion: $\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FOS}$

$$\frac{63.33}{500} + \frac{316.67}{860} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 2.02$$

8.8 Down Rod Stress Analysis

The Axial stress: $\sigma = \frac{F}{A}$, where A is the cross-sectional area.

The cross-sectional area $(A) = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}(0.15^2 - 0.08^2) = 0.01263\, m^2$

$$\sigma = \frac{F}{A} = \frac{3 \times 10^6}{0.01263} = 238\, MPa$$

The Torsional Shear stress: $\tau = \frac{Tr}{j}$

The polar moment of inertia $(J) = \frac{\pi}{32}(D^4 - d^4) = \frac{\pi}{32}(0.15^4 - 0.08^4) = 4.57 \times 10^{-5}\, m^4$

The Torque (T) has been calculated before = 107150 N.m

$$\tau = \frac{Tr}{j} = \frac{107150 \times 0.075}{5.57 \times 10^{-4}} = 175.8\, MPa$$

Using Von Mises Stress Equation:

$$\sigma_{all} = \sqrt{\sigma^2 + 3\tau^2}$$
$$= \sqrt{238^2 + 3 \times 175.8^2} = 386.2\, MPa$$

8.8.1. Down rod total deformation analysis

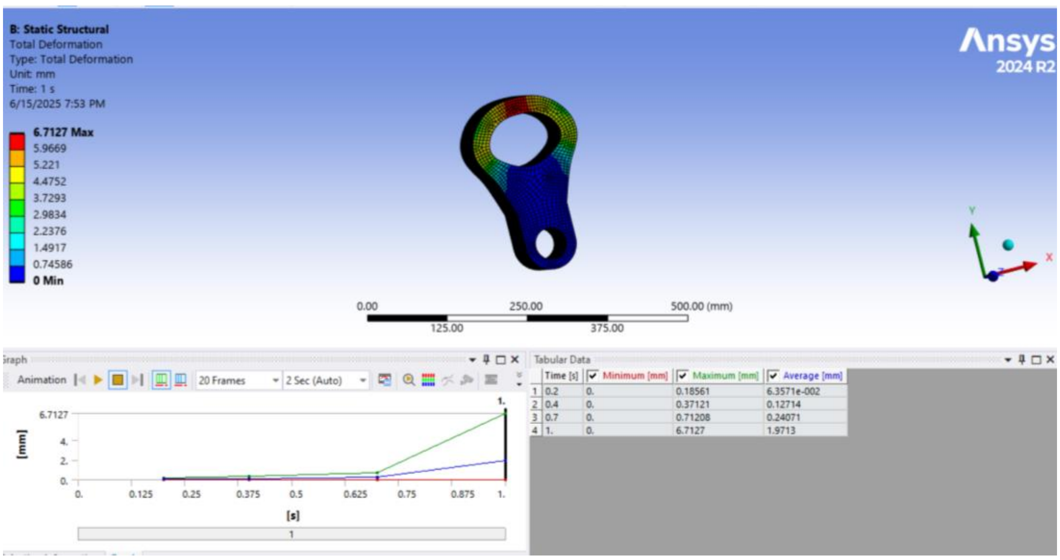


Figure 37:: Total deformation of the down rod

Deformation results were minimal, validating rigidity. The Von Mises stress and shear stress analysis confirmed that stress magnitudes remained under the allowable limit of AISI 4340 (740 MPa). Manual calculations and ANSYS results correlate well, confirming structural competence.

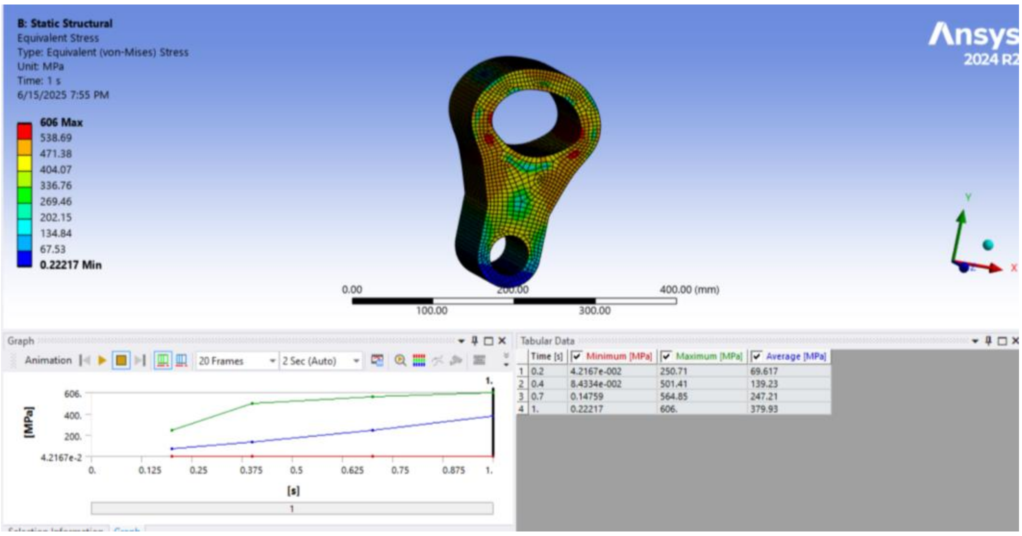


Figure 38: Equivalent (Von-Mises) stress analysis for the down rod

Part Name	Type of Analysis	Applied Forces / Moments	Boundary Conditions	Material Used
Down Rod	Static	3 MN axial, 107,150 Nm torque	Fixed bottom face	AISI 4340

Figure 39:Analysis proprieties for the down rod

8.8.2. Down Rod Safety Analysis

The allowable yield strength for it = 740 MPa and max stress acts on the part is

$$\sigma_{max} = 386.2\ MPa$$

$$FOS = \frac{\sigma_{allowable}}{\sigma_{appied}} = \frac{740}{386.2} = 1.92$$

The Min stress acts on the part is neglectable compared to its Max.

The Ultimate Tensile Strength $\sigma_u = 1080\ MPa$ for AISI 4340, the endurance limit using the equation below:

$$\sigma_e = 0.577 \times \sigma_u = 0.577 \times 1080 = 623\, MPa$$

Now using Goodman Criterion:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FOS}$$

$$\frac{64.35}{500} + \frac{322}{860} = \frac{1}{FOS} \rightarrow\rightarrow FOS = 1.78$$

8.9 Clutch and Break Analysis

We calculated the press Capacity = 300 MN**,** stroke per minute (SPM) = 15, maximum torque = 107,150 N·m, rotational Speed (ω) = 1.57 rad/s (~15 rpm)

μ (friction coef.) = 0.3 (steel-steel dry contact), factor of safety = 1.5

$$T_{design} = 1.5 \times 107150 = 16100\, N.m$$

For multi-disc clutch or break, Torque transmission (T)

$$T = n.\mu.F.R_m$$

Where:

A) T: Transmitted torque
B) N = Number of friction surfaces
C) μ: Coefficient of friction (0.3)
D) F: Normal force (N)
E) R_m : Mean radius (m)

Outer radius R = 0.25 m, and inner radius = r = 0.15 m

Uniform pressure:

$$R_m = \frac{2}{3}.\frac{R^3 - r^3}{R^2 - r^2}$$

$$R_m = \frac{2}{3}.\frac{0.25^3 - 0.15^3}{0.25^2 - 0.15^2} = 0.211\, m$$

Uniform wear:

$$R_m = \frac{R + r}{2} = \frac{0.25 + 0.15}{2} = 0.2\, m$$

Let's assume n = 6 effective surfaces (3 discs on each side), where (n) Number of Friction Surfaces:

$$F = \frac{T}{n.\mu.R_m} = \frac{160}{6 \times 0.3 \times 0.211} = 422727\, N \quad \text{[Clutch (Uniform Pressure)]}$$

The pressure on friction surface: σ

$$A = \pi(R^2 - r^2) = \pi(0.25^2 - 0.15^2) = 0.1256\, m^2$$

9- Competitive Benchmarks

To assess the viability and positioning of our Link Drive Mechanical Press in the current market, a comprehensive benchmark study was conducted. Our research compares our design with existing commercial link-motion mechanical presses used in the automotive manufacturing sector. The goal is to highlight how our system aligns with or exceeds current industry standards.

9.1 Performance vs Cost Evaluation

Our design strikes an effective balance between performance and cost. By optimizing the dimensions of key components and using advanced materials, we achieve safe stress limits while maintaining manufacturing feasibility. With a power requirement of 168.23 kW, our system is slightly more energy-efficient than some competitors. Furthermore, the average Factor of Safety (FoS) for our components ranges from 1.69 to 2.01, demonstrating a safe operating margin across all major parts. The estimated build cost is competitive at approximately $78,000, making our press a cost-effective solution for mid-to-high-volume production lines.

9.1.2. Graphical Comparison

To further illustrate the efficiency and safety of our Link Drive Mechanical Press compared to existing industry solutions, the following graphs present a visual comparison of power consumption and safety factors.

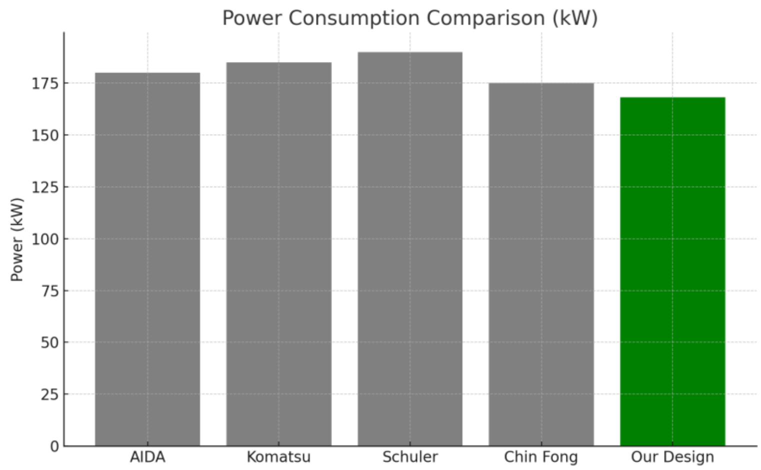


Figure 40: Power Consumption Comparison (kW)

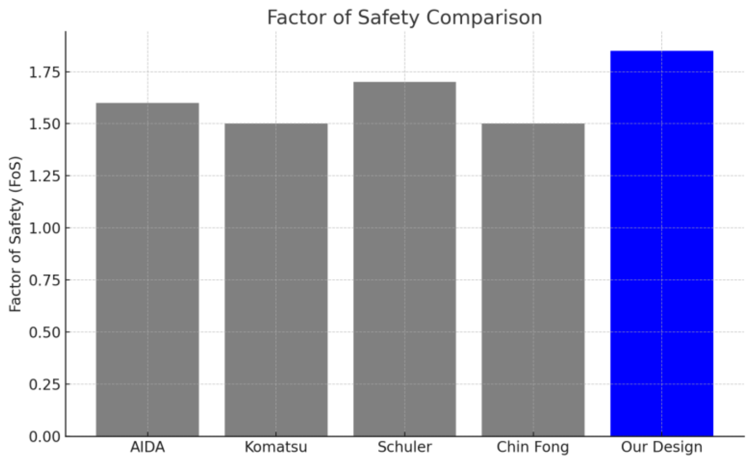


Figure 41:: Factor of Safety (FoS) Comparison

As shown in the graph, our press consumes significantly less power than competing designs, which contributes to operational cost savings and better energy efficiency.

Our press design offers one of the highest safety factors among competitors, ensuring robust performance under high-load conditions while maintaining structural integrity and operational safety.

10- Used Engineering Standards and/or Codes

ISO 1328-1:2013 — Cylindrical gears — ISO system of accuracy

AGMA 2101-D04 — Fundamental rating factors and calculation methods for involute spur and helical gear teeth

DIN 3990 — Calculation of load capacity of cylindrical gears

ISO 6336 — Calculation of load capacity of spur and helical gears

ASME B106.1M — Design of transmission shafting

ISO 281:2007 — Rolling bearings — Dynamic load ratings and rating life

OSHA 1910 Subpart O — Machinery and machine guarding (safety standard for mechanical presses)

CE Machinery Directive 2006/42/EC — Essential safety requirements for machinery

SAE J404 – Chemical Compositions of SAE Alloy Steels:

AMS 6514 – Maraging Steel 300:

AISI 4340 – High Strength Alloy Steel:

AISI 4140 – Chromium-Molybdenum Steel:

ST52 – DIN 17100 Structural Steel

11- Consideration of United Nations (UN) Sustainable Development Goals

SDG 9 — Industry, Innovation and Infrastructure:
The project contributes to resilient infrastructure and sustainable industrialization by designing a high-performance mechanical press that improves productivity and precision in automotive manufacturing.

SDG 12 — Responsible Consumption and Production:
By optimizing the design for material efficiency, durability, and reduced maintenance, the press minimizes material waste and energy consumption during operation.

SDG 8 — Decent Work and Economic Growth:
The press system is designed to meet high safety standards (OSHA, CE) and improve working conditions by reducing hazards associated with traditional press operations.

SDG 13 — Climate Action:
Through energy-efficient design and careful material selection (e.g., using durable materials that extend component life and reduce the need for frequent replacements), the project helps lower the carbon footprint associated with industrial production.

12- Impact Statement

The completion of this project demonstrates the ability to engineer and validate a structurally sound, production-ready link drive press using industry practices. By integrating simulation with material science and realistic loading, this press can be reliably manufactured, deployed, and maintained in automotive or industrial forming lines. The use of advanced alloys and high-strength steels extends the system's life span, reduces maintenance intervals, and ensures safety. Moreover, the decision framework and analysis workflow used here can serve as a transferable model for other high-load mechanical systems.

13- Conclusion

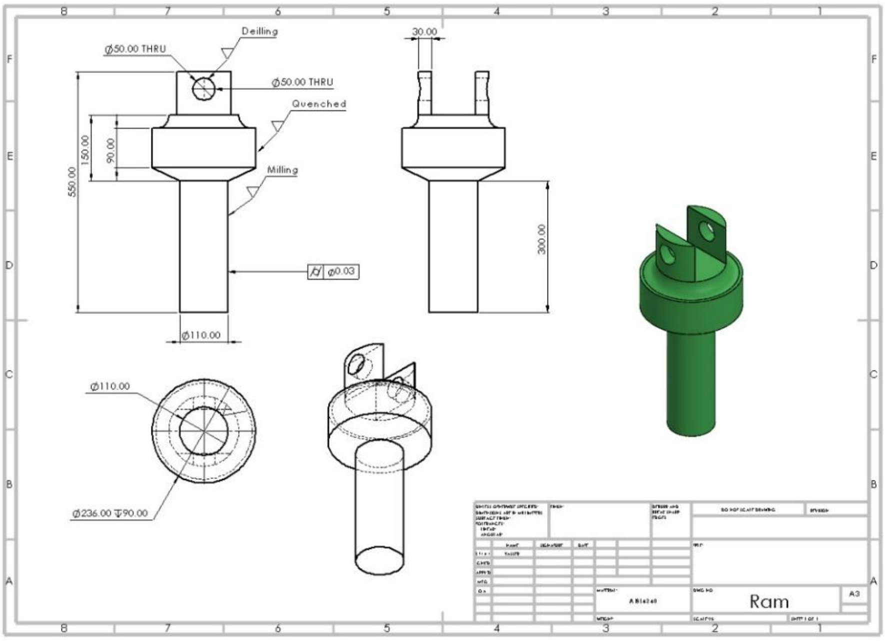
This project involved the complete mechanical design and simulation-based validation of a link drive mechanical press system. A rigorous approach was taken in both modeling and material selection to ensure the mechanical integrity, operational efficiency, and manufacturability of the system under a maximum press load of 3,000,000 N. The main gear, subjected to a torque of 107,150 Nm and a tangential force of approximately 428,600 N (calculated by $T = F \times r$), was designed using AISI 4340 due to its high strength (~850 MPa yield) and wear resistance.

Its fatigue limit and good machinability made it suitable for continuous cyclic operation. The main shaft was selected to be made of Maraging Steel 300, which offers ultra-high yield strength (~1860 MPa) and excellent fatigue performance. This choice was crucial due to the extreme loading conditions: 3,000,000 N axial compression, 107,150 Nm torque, and a 420,000 Nm bending moment (calculated using $M = F \times e$, with $e = 140$ mm). Its low thermal distortion also ensures dimensional precision under heat treatment and operation. The eccentric gear and connecting rods, both experiencing combined torsional and axial stresses, were also modeled and validated under real-world conditions. Finite element simulations confirmed that stress levels in critical areas remained below the respective yield strengths with acceptable deformation limits (<0.2 mm for shaft, <0.005 mm for base plate). Material selection was not based on strength alone. A multi-criteria decision-making approach using weighted Pugh matrices considered fatigue resistance, cost, machinability, weldability, and availability. Each matrix was backed by credible property data and referenced engineering standards such as ASTM A29 and AMS 6514. ST52 was used for the base plate to ensure stable load distribution with good weldability and cost-efficiency. Altogether, the chosen materials, coupled with validated mechanical simulations, resulted in a mechanically sound, fabrication-ready design. The system is expected to operate safely under repeated high-load cycles typical in automotive manufacturing environments.

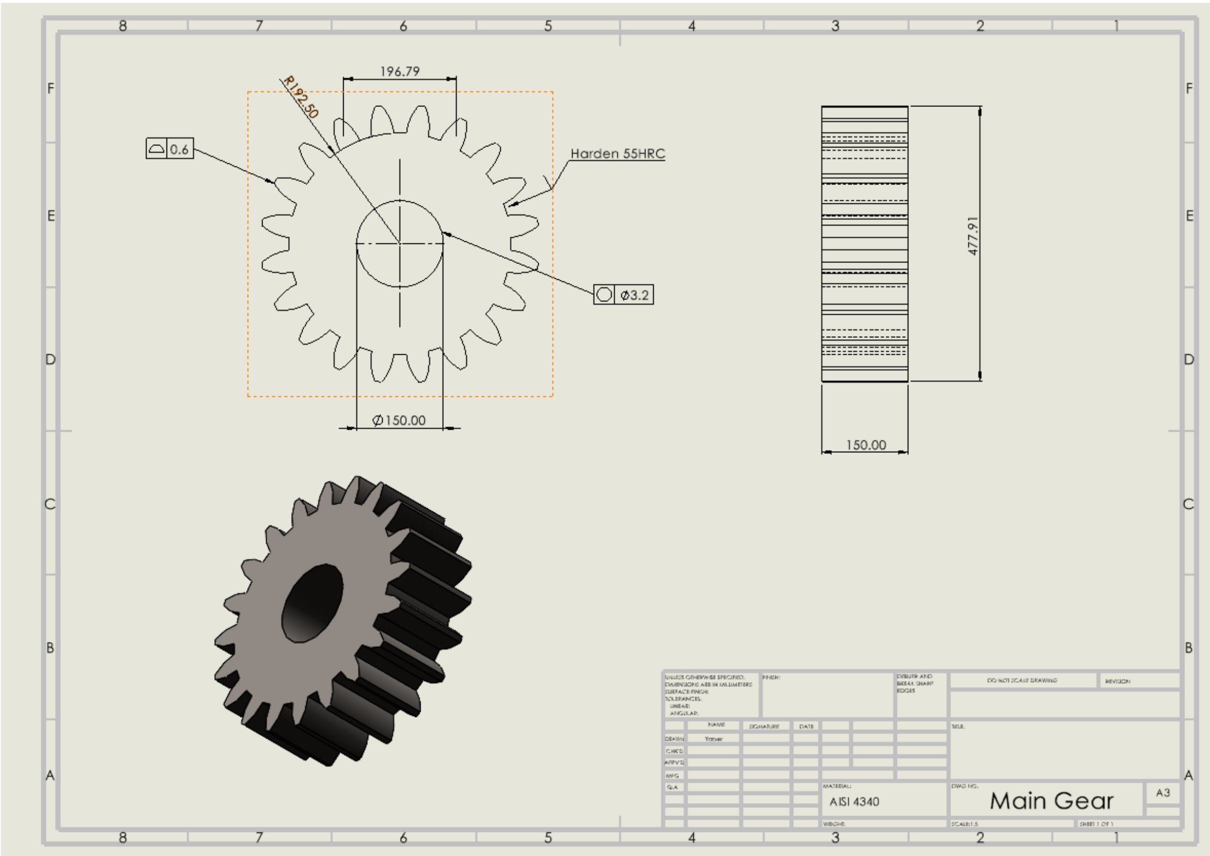
14- Appendix

14.1 Technical drawings for internal parts

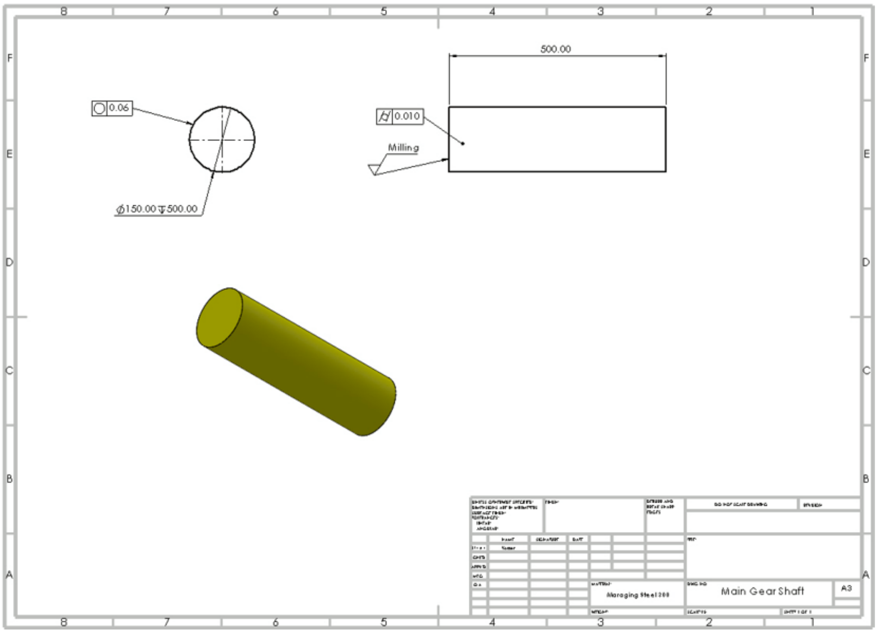
Ram:


Deilling
Ø50.00 THRU
Ø50.00 THRU
Quenched
Milling
30.00
550.00
150.00
90.00
300.00
Ø0.03
Ø110.00
Ø110.00
Ø236.00 ↧90.00
Ram
A3

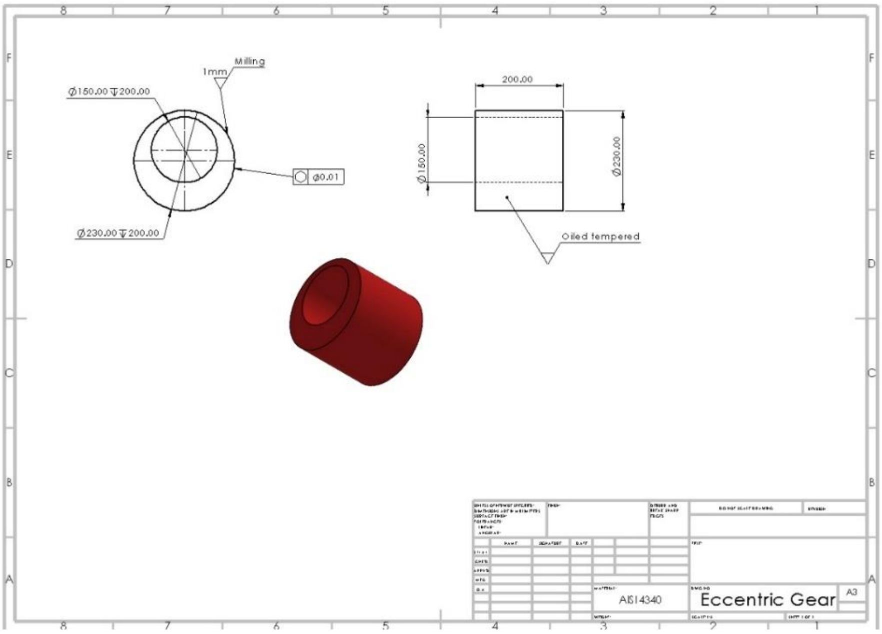
Main Gear:


196.79
R192.50
0.6
Harden 55HRC
Ø3.2
Ø150.00
477.91
150.00
AISI 4340
Main Gear
A3

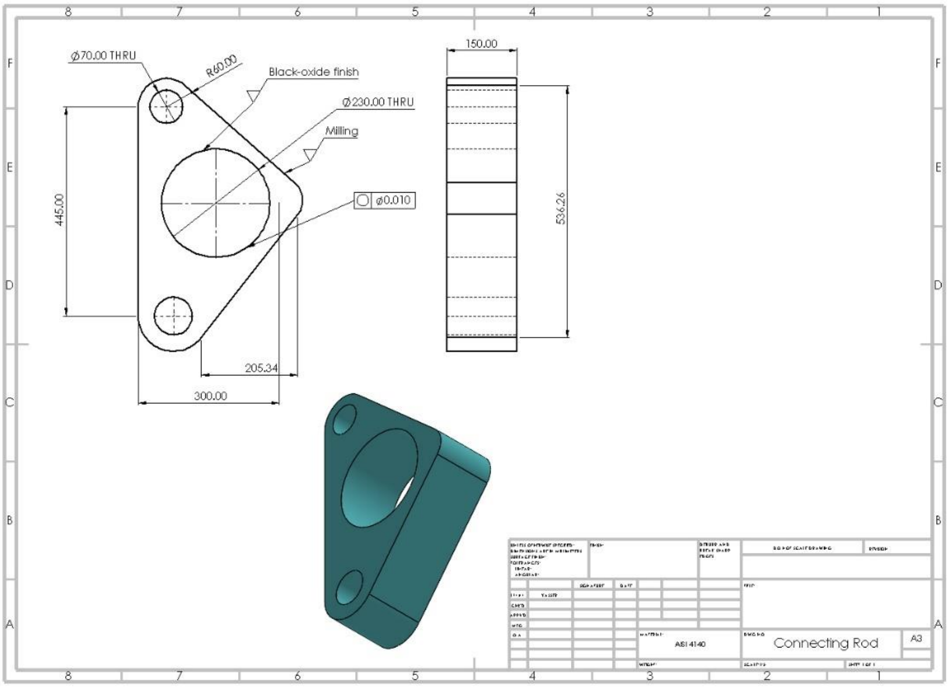
Main gear’s shaft:



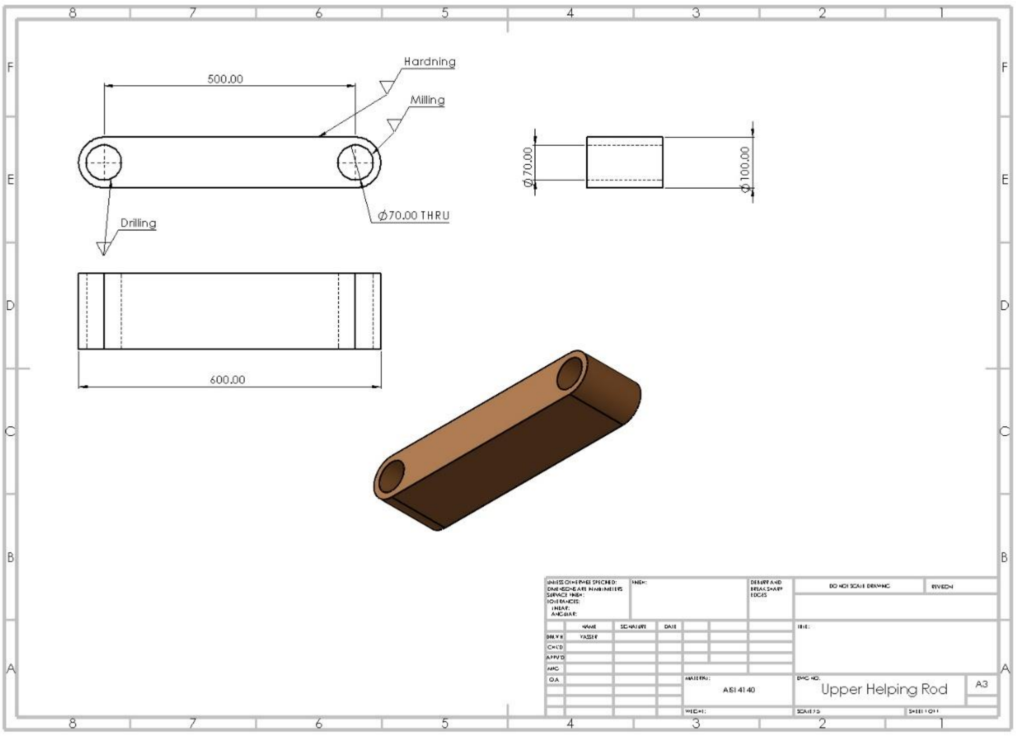
Eccentric gear:



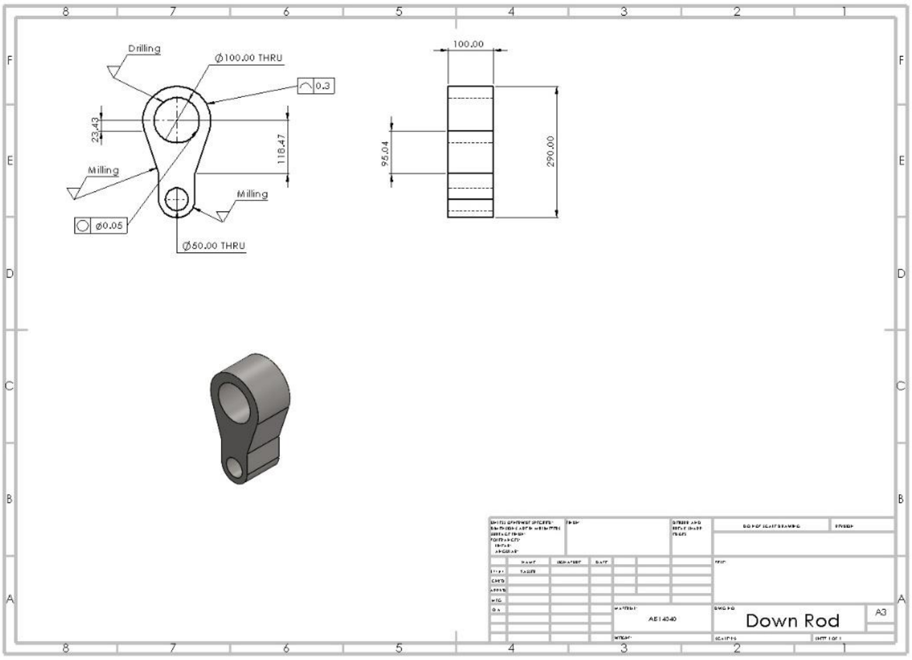
Connecting rod:



Upper helping rod:



Down rod:



14.2. MATLAB codes for stroke graph

Using **MATLAB, we** can fit a polynomial of Stroke in terms of Theta **S(**θ **).**

```
Theta = [0 30 60 90 120 150 180 210 240 270 300 330 360];
stroke = [130 121 88 46 14 2.8 0 4.6 23 60 102 125 130];
FUNCTION = polyfit(Theta, stroke,8);
theta_new = linspace(0, 360, 1000);
stroke_new = polyval(FUNCTION, theta_new);
plot(theta_new, stroke_new, 'b');
xlabel('Crank Angle (degrees)');
ylabel('Stroke (mm)');
title('Link Drive Stroke Profile (Max Stroke = 130mm)');
grid on;
xlim([0 360]);
ylim([0 140]);
xticks(0:30:360);
yticks(0:20:140);
legend('Link Drive Stroke', 'Location', 'Best');
```

Project Cost Table – Link Drive Mechanical Press Design

Item	Quantity	Material / Brand	Estimated Unit Cost (USD)	Total Cost (USD)
Main Gear	1	AISI 4340	$2,500	$2,500
Main Shaft	1	Maraging Steel 300	$3,000	$3,000
Eccentric Gear	1	AISI 4340	$1,800	$1,800
Connecting Rod	1	AISI 4140	$600	$600
Upper Helping Rod	1	AISI 4140	$450	$450
Down Rod	1	AISI 4340	$1,000	$1,000
Ram	1	AISI 4340	$1,200	$1,200
Pinion	1	AISI 4340	$600	$600
Bearings (ISO 15 ABB Series)	8	SKF/Equivalent	$300	$2,400
Motor + Gearbox	1	Siemens 1PH8 (250 kW)	$12,000	$12,000
Base Plate	1	ST52 Steel	$1,500	$1,500
Manufacturing & Machining	-	CNC, Welding, Drilling	-	$25,000
Control System & Safety	1 set	PLC, sensors, buttons	-	$5,000
Frame & Housing	1	Structural Steel	-	$10,000
Software & Simulation	-	MATLAB, ANSYS Licensing	-	$3,500
Documentation & Testing	-	Materials, Testing Equipment	-	$1,450
Total Estimated Project Cost	**$78,000 (Fits the constraints)**			

14.4 Meeting Memos

MECH/MECT4902 DESIGN PROJECT MEMO FORM

Year: 2025 **Semester: Spring**

Team No: 8

Advisor: Umut KARAGÜZEL

Project Title: Link Drive Mechanical Press Design for Automotive Industry

Meeting No: 1 **Date: 24/02/2025**

Attendants: AHMED ABUALAMRAİN **,** BARAN EKİNCİ **,** YAĞIZ ARAS

<u>Discussed Topics in the Meeting:</u>

Introduction to the project and work process;

Link drive mechanism design and the constraints of the project;

<u>Work to be Done Until Next Meeting and Responsible Team Member(s):</u>

Pressed part selection; (Team Responsible)

Link drive mechanism researches; (Team Responsible)

Task distrubution:

Ahmed: Stroke Profile

Baran: Material slection standarts

Yagiz: Kinematic analyses

Prepared by: Ahmed Abualamrain

MECH/MECT4902 DESIGN PROJECT MEMO FORM

Year: **2025** **Semester: Spring**

Team No: 8

Advisor: Umut Karagüzel

Project Title: Link Drive Mechanical Press Design for Automative Industry

Meeting No: 2 **Date: 10/03/2025**

Attendants: AHMED ABUALAMRAİN , Baran Ekinci ,
YAĞIZ ARAS , Mohammed Yasser

<u>Discussed Topics in the Meeting:</u>

Geogebra and Optimization

Stroke Profile

Vibration and Model Analysis

Material Selection

Kinematic Analysis

Gantt Chart

<u>Work to be Done Until Next Meeting and Responsible Team Member(s):</u>

Kinematic Analysis for the System Analysis

Geogebra Analysis

To research about Material Selection

To research about Vibration and Model Analysis

Mechanical analyses (Stress, strength, force)

Prepared by: BARAN EKİNCİ

MECH/MECT4902 DESIGN PROJECT MEMO FORM

Year: 2025 Semester: Spring

Team No:8…

Advisor: …………Umut Karagüzel

Project Title: …Link Drive Mechanical Press Design for Automative Industry

Meeting No: 3 Date: 27/03/2025

Attendants: AHMED ABUALAMRAİN, Baran Ekinci,

YAĞIZ ARAS, Mohammed Yasser

Discussed Topics in the Meeting:

We discussed what needs to be done regarding the Gantt chart and how it should be done.

We determined that material selection is a lengthy process and that decisions will be made separately for each component.

We talked about concept design, and it was agreed that multiple concept designs could be created.

We decided to spend one more week working on kinematic analysis.

We had a brief discussion about the shafts in link drive presses.

Work to be Done Until Next Meeting and Responsible Team Member(s):

Yağız Aras = Trying to finalize kinematic analysis

AHMED ABUALAMRAİN = Parts calcs (connecting rode, down rode, helping rode, eccentric gear)

BARAN EKİNCİ = Research about motor selection and calculations.

Mohammed Yasser= Do research on FEA analysis

Prepared by: Yağız ARAS

MECH/MECT4902 DESIGN PROJECT MEMO FORM

Year: 2025 Semester: Spring

Team No:8

Advisor: Umut Karagüzel

Project Title: …Link Drive Mechanical Press Design for Automative Industry

Meeting No: 4 Date: 29/05/2025

Attendants: AHMED ABUALAMRAİN, Baran Ekinci,

YAĞIZ ARAS, Mohammed Yasser

Discussed Topics in the Meeting:

The motor selection process was discussed.

Problems encountered during the 3D design process were discussed.

Clutch and brake calculations were discussed and information was shared.

Press body design was discussed.

Work to be Done Until Next Meeting and Responsible Team Member(s):

Yağız Aras = 3D Drawings (Press Body Design)

AHMED ABUALAMRAİN = Parts calcs (connecting rode, down rode, helping rode, eccentric gear)

BARAN EKİNCİ = Motor selection

Prepared by: Yağız ARAS

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